KHULNA UNIVERSITY OF

ENGINEERING & TECHNOLOGY

Department of Electrical and Electronic Engineering

A Project on:

Electrical Wiring Design, Installation Planning and Cost Estimation of the Ground Floor of Khan Jahan Ali Hall (South Building)

Course No: EE 2200

Course Title: Electrical Services Design-I

Submitted To:

Dr. Md. Abdur Rafiq	**Jeesun Patra Papon**
Professor	**Lecturer**
Dept. Of EEE	**Dept. Of EEE**

Submitted By: **Project Group No:10**

Serial No	Name	Roll
1	SADMAN SAKIB RADHIN	2303010
2	MAHEDI HASAN SIAM	2303025
3	MD. IMTIAZ AHMED	2303040
4	MD. NOBEL AHMED	2303055

Session:2024-2025

Year:2nd

Term:2nd

Date:08-11-2026

Index

Serial NO	Title	Page NO
01	Preface	03
02	Introduction	04
03	About This Project	05
04	Electrical Legends	06
05	AutoCAD Design	07
06	Calculation of Illumination	11
07	Number of Fan Calculation	20
08	Rated Current Calculation	22
09	Circuit Breaker Ratings	29
09	Electrical Power Flow Diagram	30
10	Conduit Length Calculation	38
11	Live Wire Calculation	52
12	Neutral Wire Calculation	69
13	ECC Wire Calculation	84
14	Estimation of Cost	92
15	Discussion	95
16	Conclusion	96
17	References	97

Preface

This project forms part of the sessional course **Electrical Service Design–1** and focuses on the practical aspects of electrical service design for buildings. The project aims to provide exposure to real-life applications of electrical wiring design, installation planning, and cost estimation.

The project addresses the **Electrical Wiring Design, Installation Planning and Cost Estimation of the Ground Floor of Khan Jahan Ali Hall (South Building)**.

The provided civil floor plan serves as the basis for preparing a complete electrical service layout in accordance with standard engineering practices. The scope of the project covers the arrangement of lighting fixtures, ceiling fans, switches, circuit breakers, sub-distribution boards, and the main distribution board. It also includes conduit routing layouts and power flow diagrams prepared using AutoCAD for clear and accurate representation.

The project incorporates essential engineering calculations to ensure safety, efficiency, and reliability of the electrical system. These calculations include illumination level assessment, rated current determination, conduit length estimation, selection of conductors based on current-carrying capacity, wire length estimation, grounding wire estimation, and selection of suitable protective devices such as appropriate circuit breakers. The project further includes a detailed cost estimation based on electrical materials, accessories, and standard installation practices.

The entire project follows the Electrical and Electronic Engineering Services for Buildings provisions of the Bangladesh National Building Code (BNBC), 2020 edition. The work progresses under the guidance and supervision of the course teachers, whose instructions ensure compliance with standards and systematic execution of the project activities.

Introduction

Electrical wiring is one of the most essential and primary concerns in the structural design of any building. Once the architect provides the blueprints, the next critical task is to prepare a precise estimation. Estimation involves determining the types, quantities, and costs of all electrical accessories to be used. This process ensures proper planning, efficient material usage, and adherence to safety and economic standards.

The major steps involved in designing the electrical layout and calculating the overall cost are as follows:

1. **COLLECT THE LAYOUT OF THE BUILDING** – Obtain the architectural drawings to understand room arrangements and spatial divisions.
2. **CREATE AN AUTOCAD MODEL OF THE BUILDING** – Develop a detailed digital model for accurate planning and visualization.
3. **MEASURE THE AREA OF DIFFERENT SECTIONS** – Determine the square footage of each room to assess illumination and load requirements.
4. **CALCULATE THE REQUIRED ILLUMINATION** – Estimate the lighting needs of each area based on purpose and standards.
5. **CALCULATE WATTAGE & AMPERAGE AND DISTRIBUTE THE LOAD** – Determine total electrical load and assign circuits to ensure safety and efficiency.
6. **DESIGN THE CONDUIT SYSTEM** – Plan the routing, dimensions, and placement of conduits for safe and organized wiring.
7. **CALCULATE THE TOTAL WIRE REQUIRED** – Estimate the length, gauge, and type of wires needed for all lighting points, switches, sockets, and panels.
8. **PERFORM COST CALCULATION** – Prepare a complete cost estimation including all materials, fittings, and labor charges.

About this Project

This project focuses on the electrical wiring design of the ground floor of Khan Jahan Ali Hall, KUET, located on the southern side of the Administration Building. The work presented here specifically addresses the wiring layout and design aspects for the first floor of the structure.

To begin with, the architectural layout is examined to provide a complete visual understanding of the building. This enables the electrical engineer to comprehend the overall structural arrangement before proceeding with the detailed design. After that, the electrical load is applied to the top-view layout to determine the distribution of power across different sections. The conduit layout—representing the path through which electrical wires will pass—is then placed on a separate structural view for clarity and precision.
Although the building serves as a hall, it contains multiple functional spaces such as rooms, washrooms, corridors, an office room, and a kitchen. Illumination requirements for each of these areas are calculated to ensure proper lighting placement and adequate luminous efficiency. Based on the total load, the full-load current is computed, which helps in selecting appropriate protective devices.

Following this, both the conduit size and the wire size are determined, along with the total wire length required for the entire floor. Finally, by considering the cost of all electrical components, conduit materials, accessories, and labor charges, the optimum overall cost for wiring the ground floor is estimated.

Electrical Legend

Sl No.	Symbol	Description
1.		MDB
2.		SDB
3.		CEILING FAN
4.		3 PIN SOCKET OUTLET
5.		LED TUBE LIGHT
6.		SURFACE CELLING LIGHT
7.		LED BULB
8.		SWITCH BOARD
9.		EARTH ELECTRODE
10.		CIRCUIT BREAKER
11.		Neutral Wire
12.		ECC Wire
13.		SWITCH BOARD to Appliances
14.		SWITCH BOARD to Appliance CONDUIT
15.		SDB to SWITCH BOARD CONDUIT
16.		MDB to SDB CONDUIT

Architectural Layout

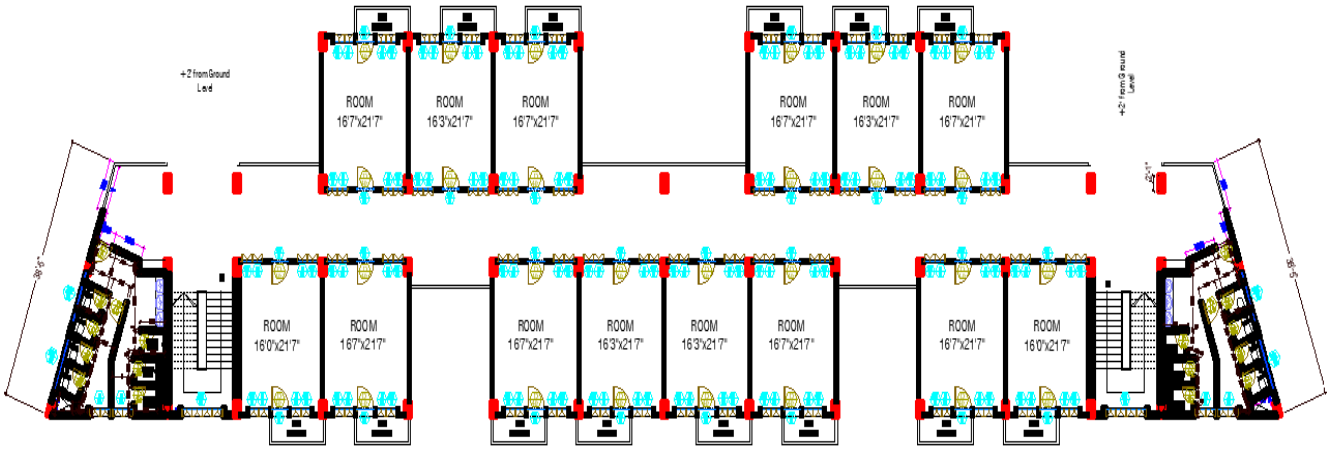


Fig-01: Floor Plan for Ground Floor of Khan Jahan Ali Hall

Electrical Fixture and Load Layout

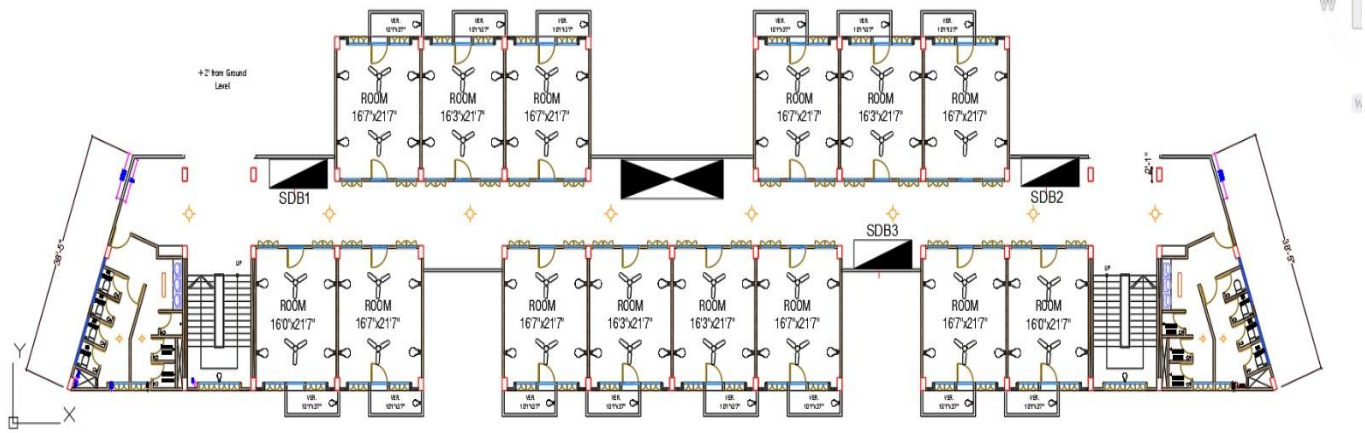


Fig-02: Electrical Equipment used in Floor Plan

Conduit Layout

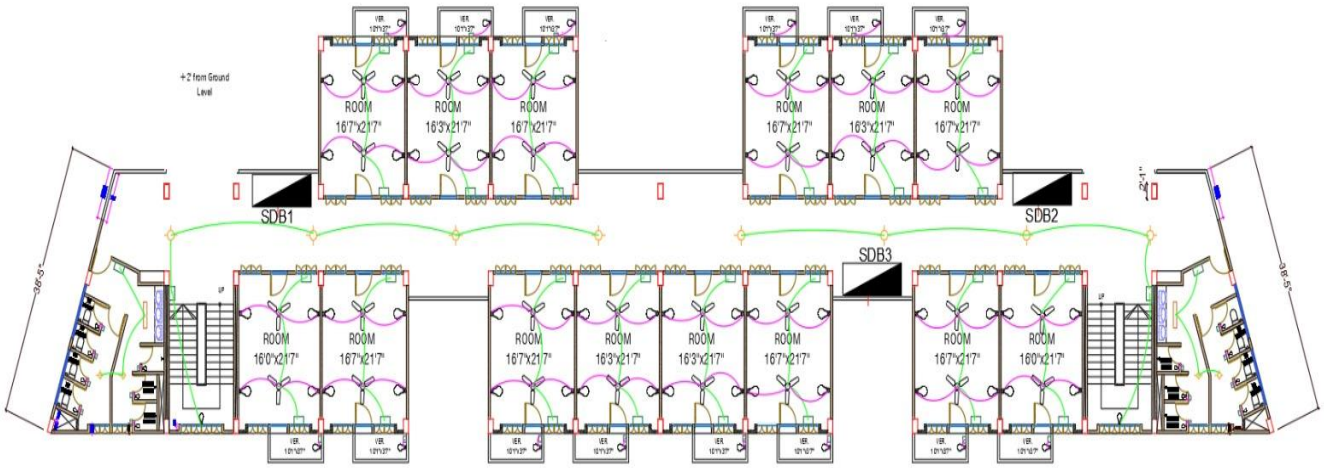


Fig-03: Conduit Layout from Switchboard to Electrical Appliance (Light &Fan).

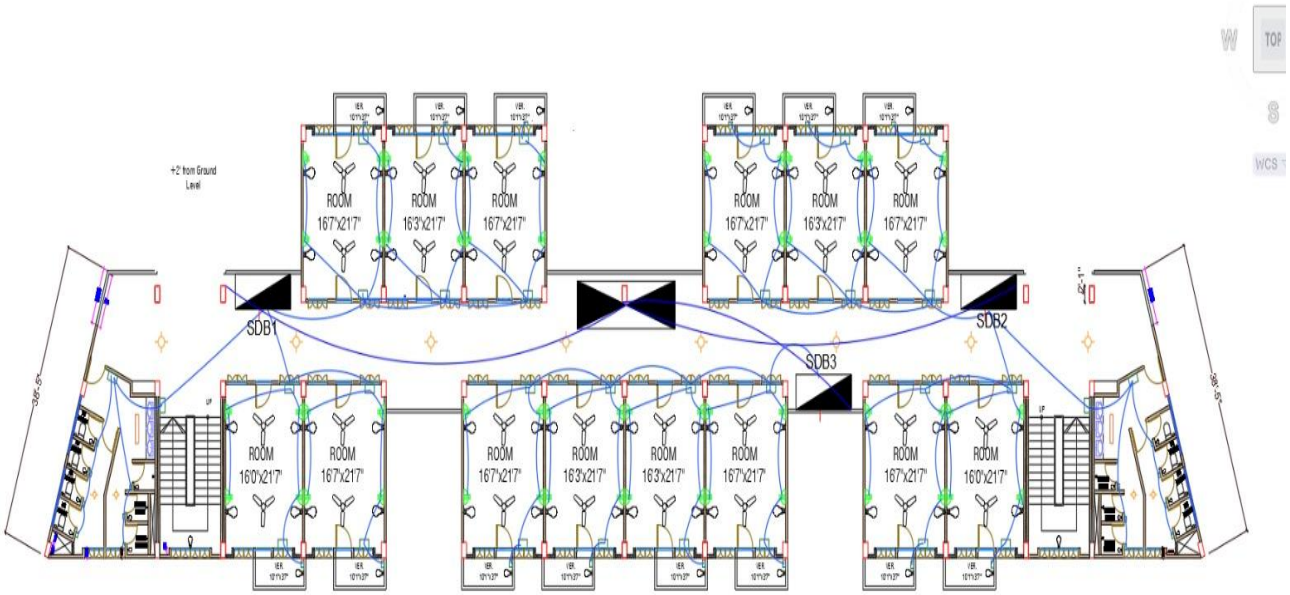


Fig-04: Conduit Layout from Sub Distribution board to Switch Board and Power Socket.

Wiring Layout

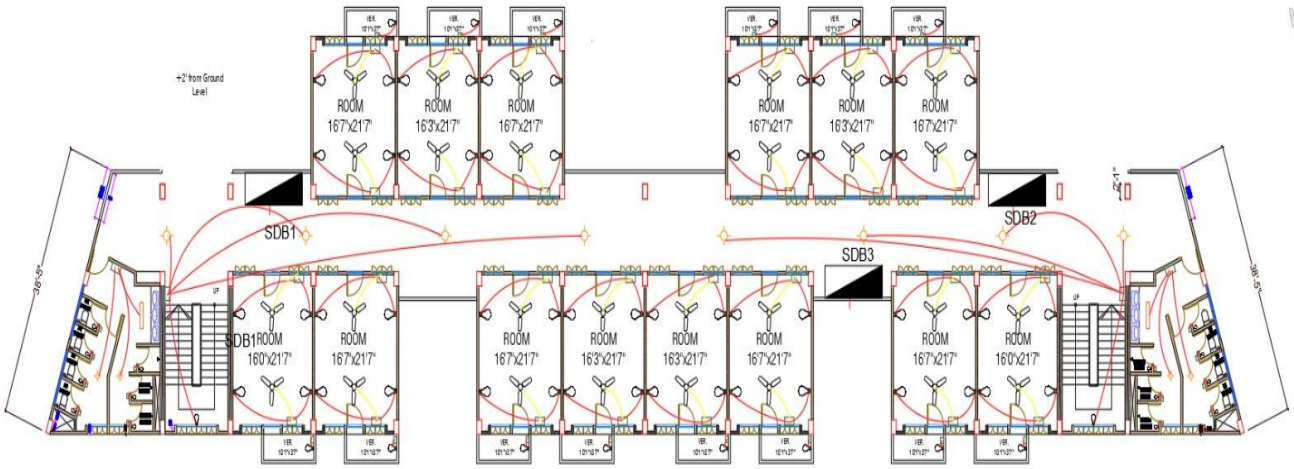


Fig-05: Live Wire from Switchboard to Electrical Appliance (Light&Fan)

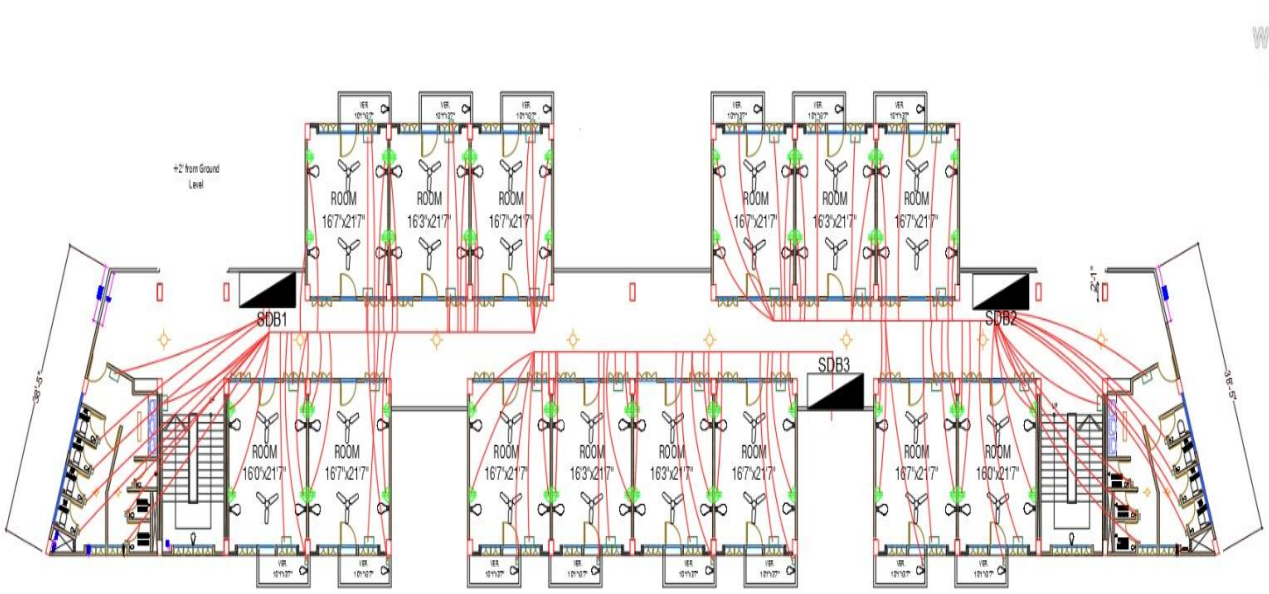


Fig-06: Live Wire from Sub Distribution board to Switch Board and Power Socket.

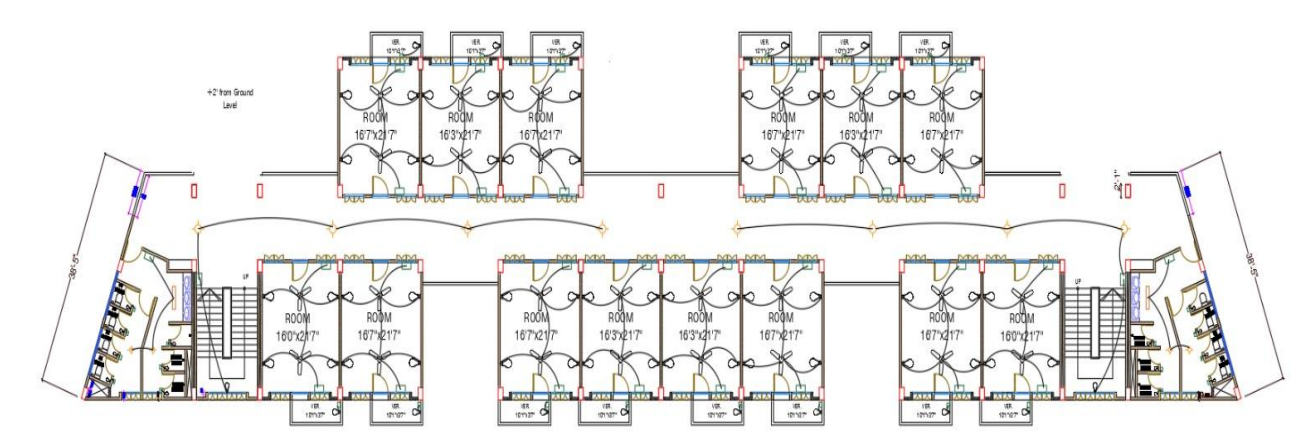


Fig-07: Neutral Wire from Switchboard to Electrical Appliance

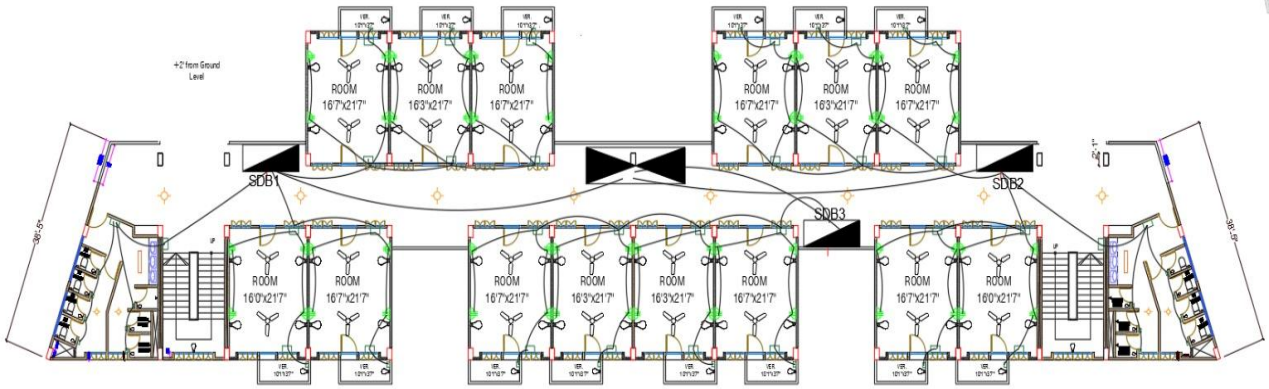


Fig-08: Neutral Wire from Sub Distribution board to Switch Board and Power Socket.

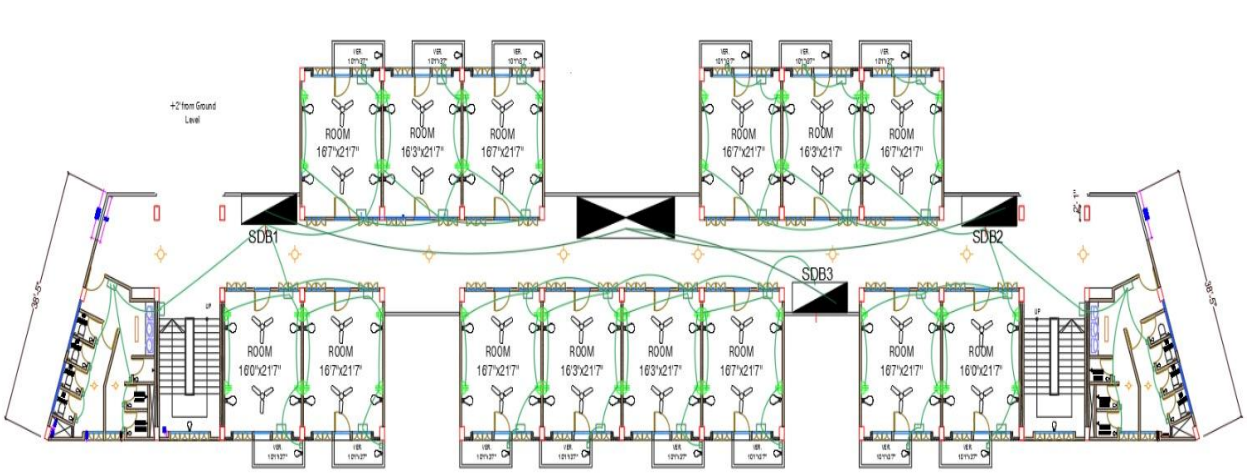


Fig-09: ECC wiring

Calculation of Illumination

Formula:

$$\text{Required illumination} = \frac{\text{Area} \times \text{Standard Illumination}}{\text{Depreciation Factor} \times \text{Utilization Factor}}$$

Depreciation factor was taken 0.8.

Utilization factor was taken as 0.7.

For Room:

Room-1:(2)

According to BNBC, the illumination =6.5 lumens/ft^2

Area=16'×21'7" ft^2 =345.33ft^2

So,

$$\text{Lumen Required} = \frac{6.5 \times 345.33}{0.7 \times 0.8}$$

= 4008.295 lumens

= 4008 lumens

Total wattage = 4008/95 watt

=42.19 watt

Length wise placement: 2 bulb

Length wise placing = (21.583/2) = 10.792ft

Mounted Height = 10ft

Length wise space height ratio = (10.792/10)

= 1.0792

Width wise placement = 2 bulb

Width wise spacing = 16ft

Width wise space height ration = (16/16) = 1

So, required light for each room:

Name	Rating	Lumens	Quantity
LED Bulb light	12W	4560	04

Room 2: (8)

According to BNBC, the illumination =6.5 lumens/ft^2

Area = 16'7''×21'7'' ft^2=358ft^2

Lumen required = $\frac{6.5\times358}{0.7\times0.8}$ =4155 lumens

Total wattage = (4155/95) watt =43.73 watt

Length wise placement: 2 bulb

Length wise placing = (21.583/2) = 10.792ft

Mounted Height = 10ft

Length wise space height ratio = (10.792/10)

= 1.0792

Width wise placement = 2 bulb

Width wise spacing = 16.583ft

Width wise space height ration = (16.583/16.583) = 1

So, required light for each room:

Name	Rating	Lumens	Quantity
LED Bulb light	12W	4560	04

Room 3:(4)

According to BNBC, the illumination =6.5 lumens/ft^2

Area: 16'3''×21'7" = 350.73 ft^2

Lumen required = $\frac{6.5\times350.73}{0.7\times0.8}$ = 4070.97 lumens

Total wattage = (4071/95)watt =42.85watt

Length wise placement: 2 bulb

Length wise placing = (21.583/2) = 10.792ft

Mounted Height = 10ft

Length wise space height ratio = (10.792/10)

= 1.0792

Width wise placement = 2 bulb

Width wise spacing = 16.25ft

Width wise space height ration = (16.25/16.25) = 1

So, required light for each room:

Name	Rating	Lumen (Lm)	Quantity
LED Bulb	12 W	4560	04

For Shower1:

According to BNBC, the illumination =9.29 lumens/ft^2

Depreciation factor was taken 0.7(Damp wall)

Utilization factor was taken as 0.7

Area=4'4"×3'6" ft^2 =15.167ft^2

Lumen Required = $\frac{15.167\times9.29}{0.7\times0.7}$ =287lumens

Total wattage = (287/100) watt =2.87watt

So, required light:

Name	Rating	Lumen	Quantity
LED Bulb	3W	300	01

For Shower 2:

According to BNBC, the illumination =9.29 lumens/ft^2

Depreciation factor was taken 0.7(Damp wall)

Utilization factor was taken as 0.7

Area=4’4”×3’10” ft^2 =16.61ft^2

Lumen Required =$\frac{16.61\times9.29}{0.7\times0.7}$ =314.933 lumens = 315 lumens

Total wattage = (315/100) watt =3.15watt

So, required light:

Name	Ratings	Lumens(lm)	Quantity
LED Bulb	5W	500	01

Basin Area:

According to BNBC, the illumination =27.87 lumens/ft^2

Depreciation factor was taken 0.7(Damp wall)

Utilization factor was taken as 0.7

Area=4’9”×7’11” ft^2

=37.6042ft^2

Lumen Required = $\frac{37.6042\times27.87.}{0.7\times0.7}$ =2138.83lumens = 2139 lumens

Total wattage = (2139/110) watt =19.44watt

Name	Rating	Lumen	Quantity
LED Tube Light	20W	2200	01

Washroom 1:

According to BNBC, the illumination = 9.29lumens/ft^2

Depreciation factor was taken 0.7(Damp wall)

Utilization factor was taken as 0.7

Area=4'4"×4' ft^2 =17.33ft^2

Lumen Required = $\frac{17.33\times9.29.}{0.7\times0.7}$ =328.626 lumens = 329 lumens

Total wattage = (329/110) watt =2.987watt

Name	Rating	Lumen	Quantity
LED Bulb	3W	330	01

Washroom 2:

According to BNBC, the illumination =9.29 lumens/ft^2

Depreciation factor was taken 0.7(Damp wall)

Utilization factor was taken as 0.7

Area=4'4"×3'7 ft^2 =15.53ft^2

Lumen Required = $\frac{15.53\times9.29.}{0.7\times0.7}$ = 294.4lumens = 295 lumens

Total wattage = (295/100)watt =2.95watt

Name	Ratings	Lumen	Quantity
LED Bulb	3W	300	01

For Storeroom:

According to BNBC, the illumination =9.29 lumens/ft^2

Depreciation factor was taken 0.8

Utilization factor was taken as 0.7

Area=13.45ft^2

Lumen Required $= \frac{9.29 \times 13.45}{0.7 \times 0.8}$ = 223.13lumens = 224 lumens

Total wattage = (224/100)watt =2.24watt

Name	Ratings	Lumen	Quantity
LED Bulb	3W	300	01

For Corridor:

According to BNBC, the illumination =9.29 lumens/ft^2

Area=2643.0389 ft^2

Lumen Required = $\frac{9.29 \times 2643.0389}{0.7 \times 0.8}$ = 43846.127 lumens

Total wattage =43846.127 / 110 = 398.6 W

So, required light:

Name	Rating	Lumen (Lm)	Quantity
Surface ceiling light	50 W	44000	08

For Staircase:

According to BNBC, the illumination =9.29 lumens/ft^2
Area = 290 ft^2

So,

Lumen Required $= \frac{9.29 \times 290}{0.7 \times 0.8}$

= 4810.89 lumens

Total wattage = 4810/10= 48 W

So, required light:

Name	Ratings	Lumen(lm)	Quantity
LED Bulb	50W	5000	01

For Balconies:

According to BNBC, the illumination =9.29 lumens/ft^2
Area = 10'1"×3'7" ft^2 =36.132 ft^2

So,

Lumen Required $= \frac{9.29 \times 36.131}{0.7 \times 0.8}$

= 600lumen

Total wattage = 600/ 100

=6 W

So, required light:

Name	Rating	Lumen	Quantity
Solar LED Bulb	6W	600	01

For Open space:

According to BNBC, the illumination =9.29 lumens/ft^2
Area = 190 ft^2

So,

Lumen Required $= \frac{9.29 \times 190}{0.7 \times 0.8}$ = 3151.96 lumen =3152 lumen

Total wattage = 3152/ 95=33.18 W

So, required light:

Name	Rating	Lumen	Quantity
Surface ceiling Light	20W	3800	02

Number of Fan Calculation

The spacing between ceiling fans is around 10ft in horizontal plane and 11ft in vertical plane

The ceiling fans are 56", 50 Hz, 310 rpm and 100 W.

Required Fan for Room-1:

Length = 21’7” = 21.583’

So, lengthwise required fan = (21.583 /11) = 1.96 ≈ 2 pcs

Width = 16’

So, widthwise required fan = (16/ 12) = 1.2 ≈ 1 pcs

Required Fan for Room-2:

Length = 21’7” = 21.583’

So, lengthwise required fan = (21.583 /11) = 1.96 ≈ 2 pcs

Width = 16’7” = 16.583’

So, widthwise required fan = (16.583/ 12) = 1.28 ≈ 1 pcs

Required Fan for Room-3:

Length = 21’7’’ = 21.583’

So, lengthwise required fan = (21.583 /11) = 1.96 ≈ 2 pcs

Width = 16’3’’ = 16.25’

So, widthwise required fan = (16.25/ 12) = 1.25 ≈ 1 pc

Summary of Fan Requirements

Place	Rating (W)	Quantity	Wattage (W)
Room 1	100	4	400
Room 2	100	16	1600
Room 3	100	8	800
TOTAL		28	2800

Rated Current Calculation

- The Line to Neutral Voltage is, V_P = 230 volts.
- The Line-to-Line voltage is, V_L =400V
- Assuming, Power Factor, cos Θ = 0.9, Safety Multiplier=1.65
- Including 20% additional load, the load current is, $I = \frac{P}{1.732 * VL \cos \theta}$

For each Power socket, Full load current is 13 A,

Wattage per Socket=230V*13A=2990 watt

For Sub distribution Board-01(SDB-01):

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage (watt)	Total Current (amperes)
1	LED Bulb	20	12W	1	240	1.16
2	Ceiling Fan	10	100W	1.65	1650	7.97
3	LED Bulb	4	3W	0.8	9.6	0.046
4	Solar LED Bulb	5	6W	0.75	22.5	0.11
5	Surface Ceiling Light	4	50W	0.7	140	0.676
6	Surface Ceiling Light	2	20W	0.7	28	0.135
7	LED Bulb	1	5W	0.8	4	0.02

8	LED Tube light	1	20W	0.9	18	0.087
9	LED Bulb	3	3W	0.75	6.75	0.0326
10	LED Bulb	01	50W	0.8	40	0.19
11	3 Pin Socket 13A	20	2990W	0.5	29900	144.44
Total connected Electrical Load					32058.85	154.87

Including 20% additional load,

The total load is=32058.85+0.20*32058.85=38470.62watt.

Total Three-Phase Current for SDB-01: I= $\frac{P}{1.732*V*cos\theta}$= $\frac{38470.62}{1.732*400*0.9}$ =61.7A

For Sub distribution Board-02(SDB-02):

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage (watt)	Total Current (amperes)
1	LED Bulb	20	12W	1	240	1.16
2	Ceiling Fan	10	100W	1.65	1650	7.97
3	LED Bulb	4	3W	0.8	9.6	0.046

4	Solar LED Bulb	5	6W	0.75	22.5	0.11
5	Surface Ceiling Light	4	50W	0.7	140	0.676
6	Surface Ceiling Light	2	20W	0.7	28	0.135
7	LED Bulb	1	5W	0.8	4	0.02
8	LED Tube light	1	20W	0.9	18	0.087
9	LED Bulb	3	3W	0.75	6.75	0.0326
10	LED Bulb	01	50W	0.8	40	0.19
11	3 Pin Socket 13A	20	2990W	0.5	29900	144.44
Total connected Electrical Load					32058.85	154.87

Including 20% additional load,

The total load is=32058.85+0.20*32058.85=38470.62watt.

Total Three-Phase Current for SDB02: I= $\frac{P}{1.732*Vcos\theta}$= $\frac{38470.62}{1.732*400*0.9}$ =61.7A

For Sub distribution Board-03(SDB-03):

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor	Total Wattage (watt)	Total Current (amperes)
1	LED Bulb	16	12W	1	192	0.93
2	Ceiling Fan	8	100W	1.65	1320	6.38
3	3 pin socket 13A	16	2990W	0.5	23920	115.55
4	Solar LED Bulb	4	6W	0.75	18	0.0869
					25450	122.95

Including 20% additional load,

The total load is=25450+0.20*25450=30540watt.

Total Three-Phase Current for SDB03: I= $\frac{P}{1.732*V cos\theta}$= $\frac{30540}{1.732*400*0.9}$ =48.97A

Total Load from 3 SDB=38470.62*2+30540=107481.24 watt

Total three phase Main Distribution Board Current=$\frac{107481.24}{1.732*400*0.9}$ =172.38A

FROM SUBDISTRIBUTION BOARD TO SWITCH BOARD:

For Bedrooms: (all 14 rooms)

For SwitchBoard-1(SB1):

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage(W)	Total Current(A)
1	LED Bulb	02	12	1	24	0.116
2	Ceiling Fan	01	100	1.65	165	0.797
Total Connected Load					189	0.92

Considering 20% overload,

The rated current of SwitchBoard-01 is =0.92*1.2=1.10 A

For SwitchBoard-2(SB2):

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage(W)	Total Current(A)
1	LED Bulb	02	12	1	24	0.116
2	Ceiling Fan	01	100	1.65	165	0.797
Total Connected Load					189	0.92

Considering 20% overload,

The rated current of SwitchBoard-01 is =0.92*1.2=1.10 A

For Shower-01: For Switch Board-29,20(SB29, SB30):

Per Switchboard:

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage(W)	Total Current(A)
1	LED Bulb	01	3	0.75	2.25	0.0108

Considering 20% overload,

The rated current of Switchboard is =0.0108*1.2=0.01296 A

For shower-02: For Switch Board-31(SB31):

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage(W)	Total Current(A)
1	LED Bulb	01	5	0.75	3.75	0.018

Considering 20% overload, the rated current is=0.018*1.2=0.0216A

For Basin Area:

For Switch Board-32(SB32):

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage(W)	Total Current(A)
1	LED Tube light	01	20	0.9	18	0.0978
2	Surface Ceiling Light	02	20	0.7	28	0.135
Total Connected Electrical Load					46	0.233

Considering 20% overload, the rated current is=0.233*1.2=0.2796A

For Per washroom:

Switch Board-33(SD33)

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage(W)	Total Current(A)
1	LED Bulb	01	3	0.75	2.25	0.01087

Considering 20% overload, the rated current is=0.018*1.2=0.0216A

For Corridor: For Switchboard34,35(SB-34,SB-35)

Per switchboard:

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage(W)	Total Current(A)
1	Surface Ceiling Light	04	50	0.7	140	0.676
2	LED Bulb	01	50	0.8	40	0.193
Total Connected Electrical Load					180W	0.869A

Considering 20% overload, the rated current is=0.869*1.2=1.0428A

For Balconies: Total Switch Board number is 14.

Per Switchboard (SB):

Serial No	Appliance Name	Quantity	Watt per component	Diversity Factor/safety multiplier	Total Wattage(W)	Total Current(A)
1	Solar LED Bulb	01	6	0.75	4.5	0.02174

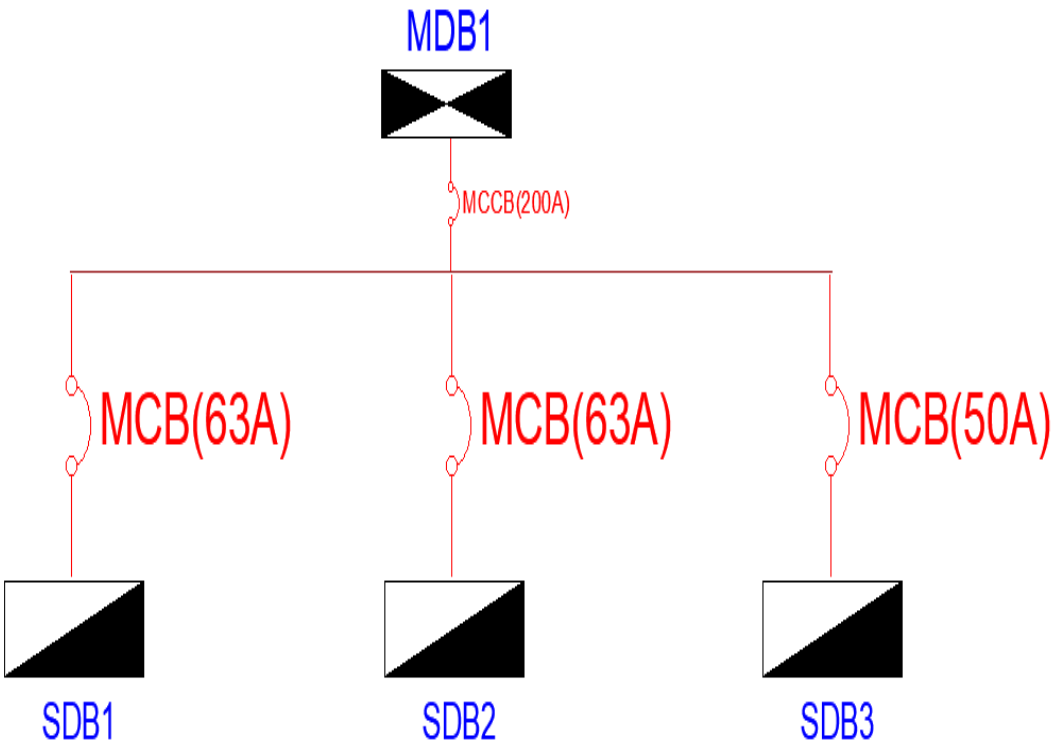
Considering 20% overload, the rated current is=0.02174*1.2=0.026A

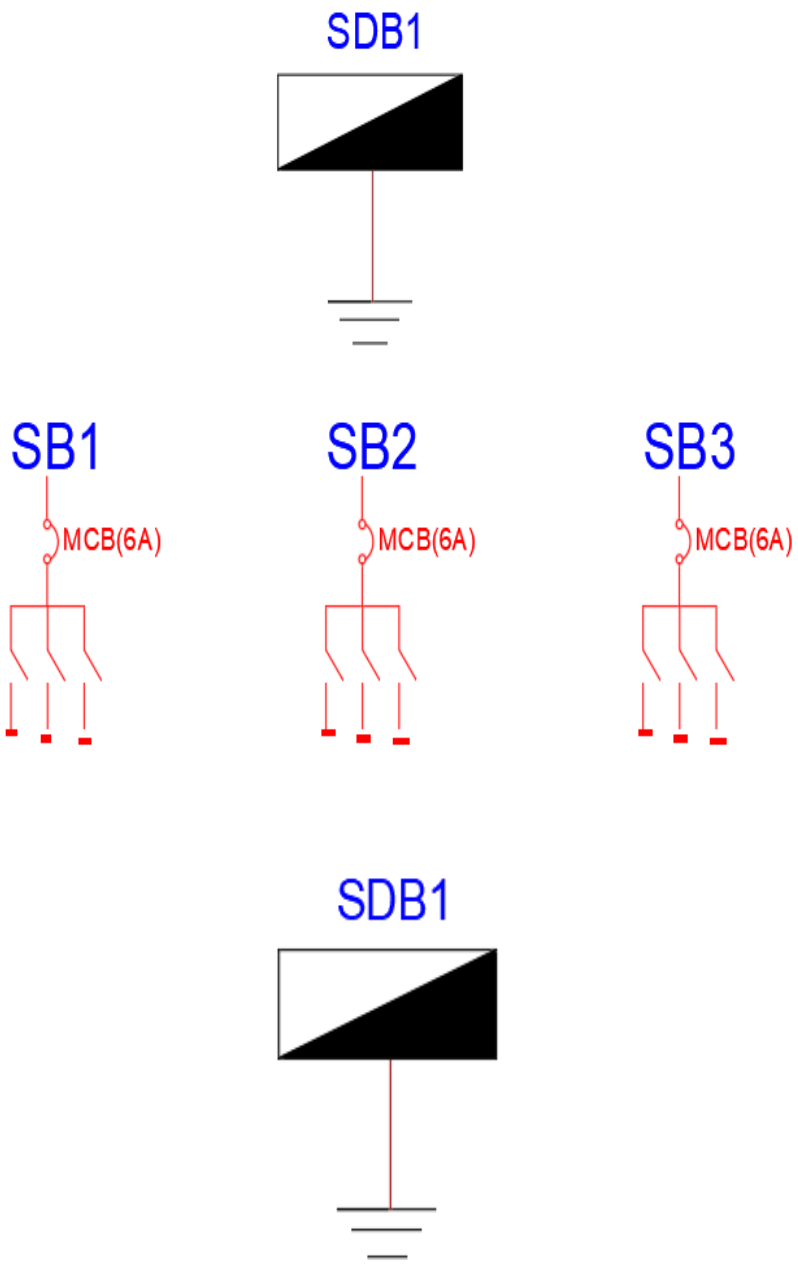
Circuit Breaker Ratings

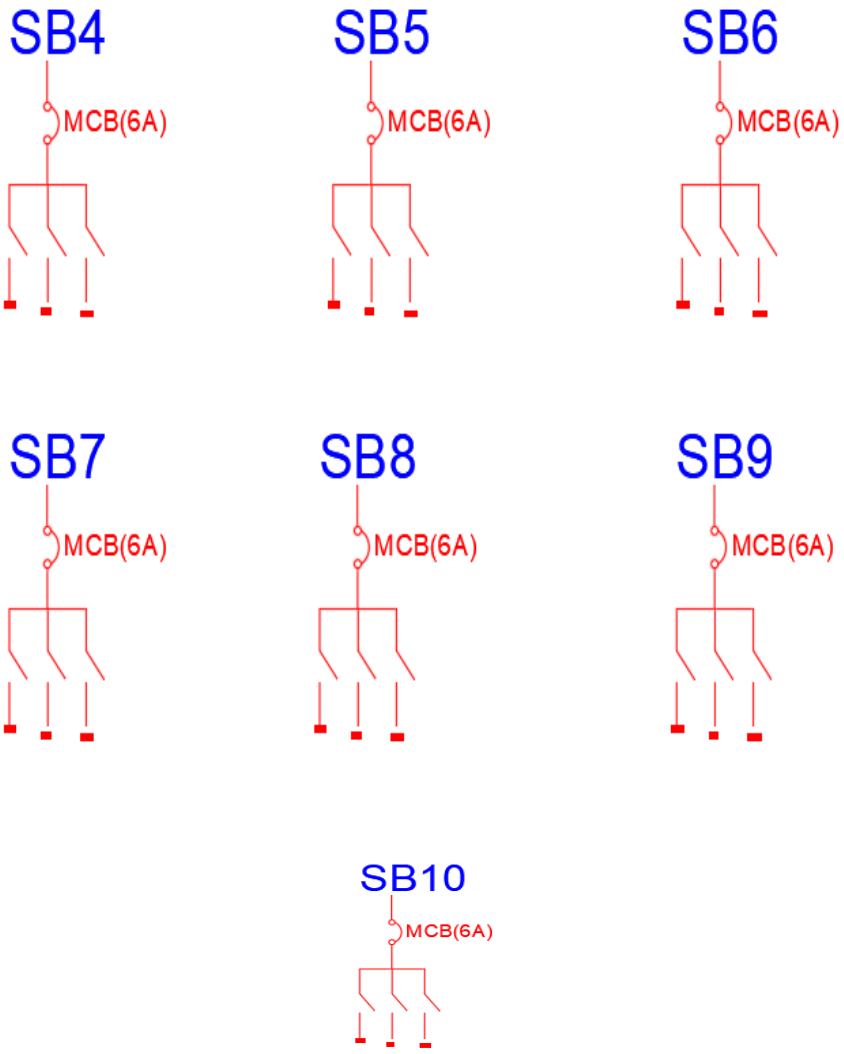
Area/Space	Circuit Breaker for Switchboard /3 pin Socket Socket	Rated Current(A) (considering Safety factor)	Circuit Breaker Rating(A)
Per Room (Total14)	SB-01	1.10	6
	SB-02	1.10	6
Per Balcony (14)	Per SB	0.026	6
Washroom	Per SB	0.0216	6
Shower-1	SB	0.01296	6
Shower-2	SB	0.0216	6
Corridor& Staircase	SB	1.0428	6
Basin Area	SB	0.2796	6
Room	Per 3Pin Socket	17.33	20
SDB-1		61.7	63
SDB-2		61.7	63
SDB-3		48.97	50
MDB		172.38	200A MCCB

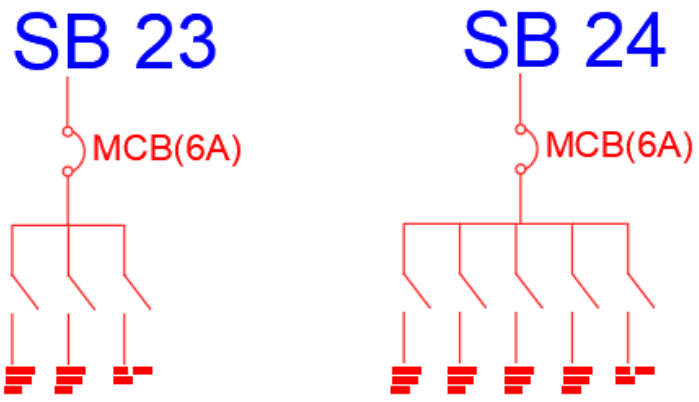
Electrical Power Flow Diagrams

Using Circuit Breaker Ratings




SDB1
SB1
MCB(6A)
SB2
MCB(6A)
SB3
MCB(6A)
SDB1

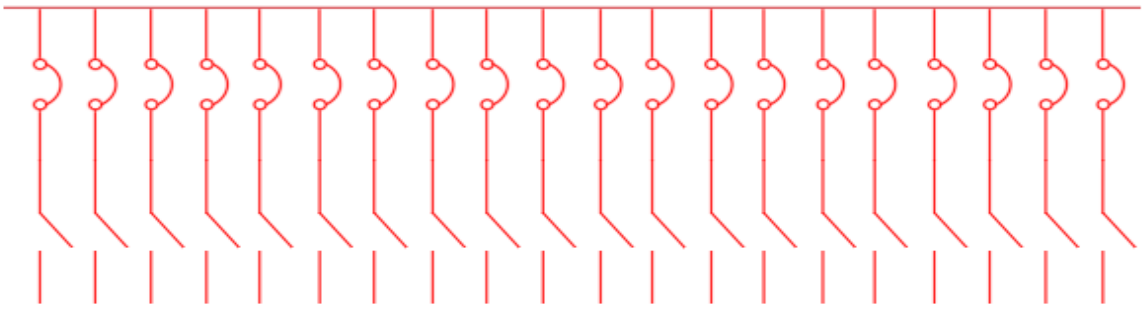

SB4
MCB(6A)
SB5
MCB(6A)
SB6
MCB(6A)
SB7
MCB(6A)
SB8
MCB(6A)
SB9
MCB(6A)
SB10
MCB(6A)
MCB(6A)
SB 11
SB 12
SB 13
SB 14
SB 15
MCB(6A)
SB 16
SB 17
SB 18
SB 19
Washroom
MCB(6A)
SB 20
SB 21
SB 22
Shower

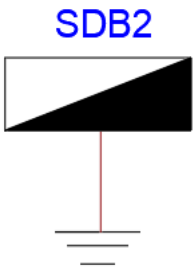


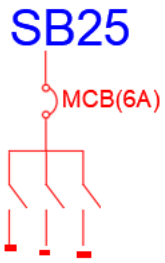
Open Space

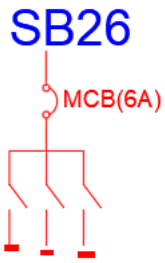
MCB(20A)

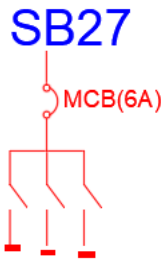
3pin Socket

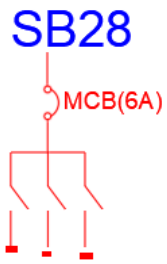


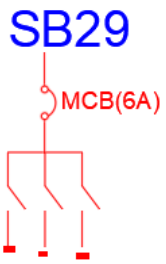

SDB2

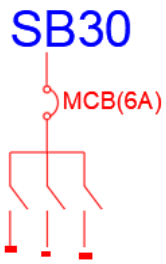

SB25
MCB(6A)

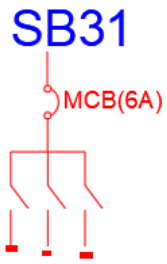

SB26
MCB(6A)


SB27
MCB(6A)

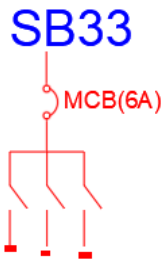

SB28
MCB(6A)

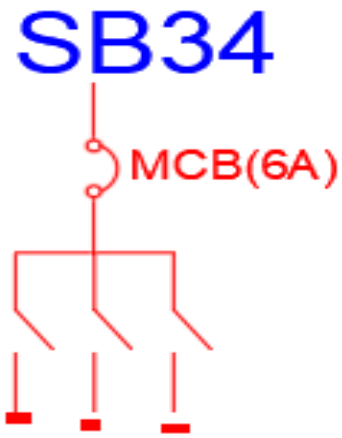

SB29
MCB(6A)

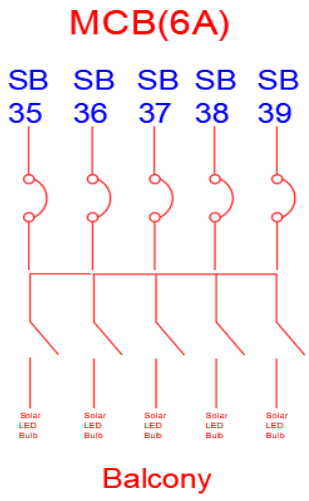

SB30
MCB(6A)

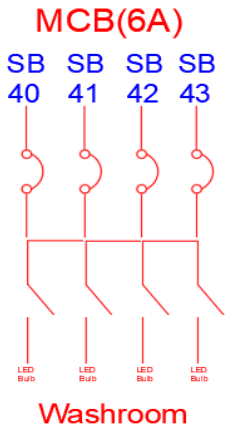

SB31
MCB(6A)

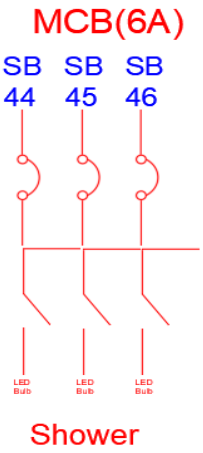

SB32
MCB(6A)

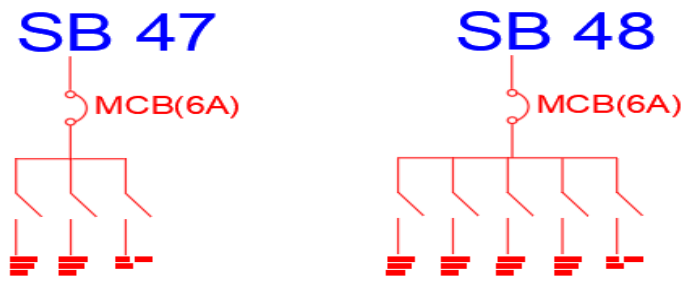

SB33
MCB(6A)


SB34
MCB(6A)


MCB(6A)
SB 35
SB 36
SB 37
SB 38
SB 39
Solar LED Bulb
Balcony


MCB(6A)
SB 40
SB 41
SB 42
SB 43
LED Bulb
Washroom

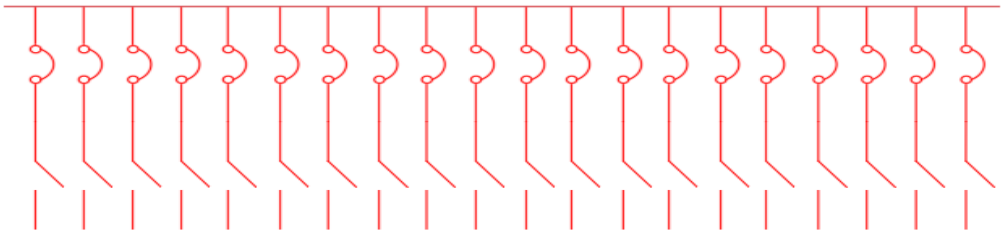

MCB(6A)
SB 44
SB 45
SB 46
LED Bulb
Shower

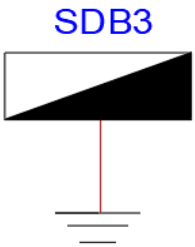


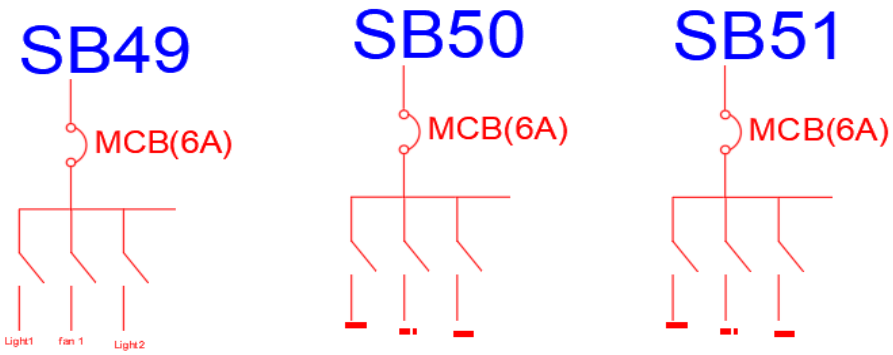
Open Space

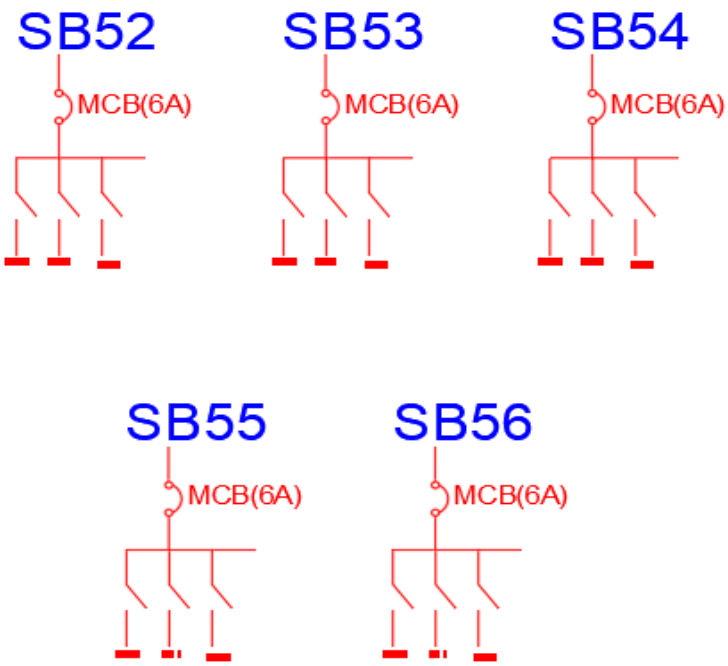
MCB(20A)

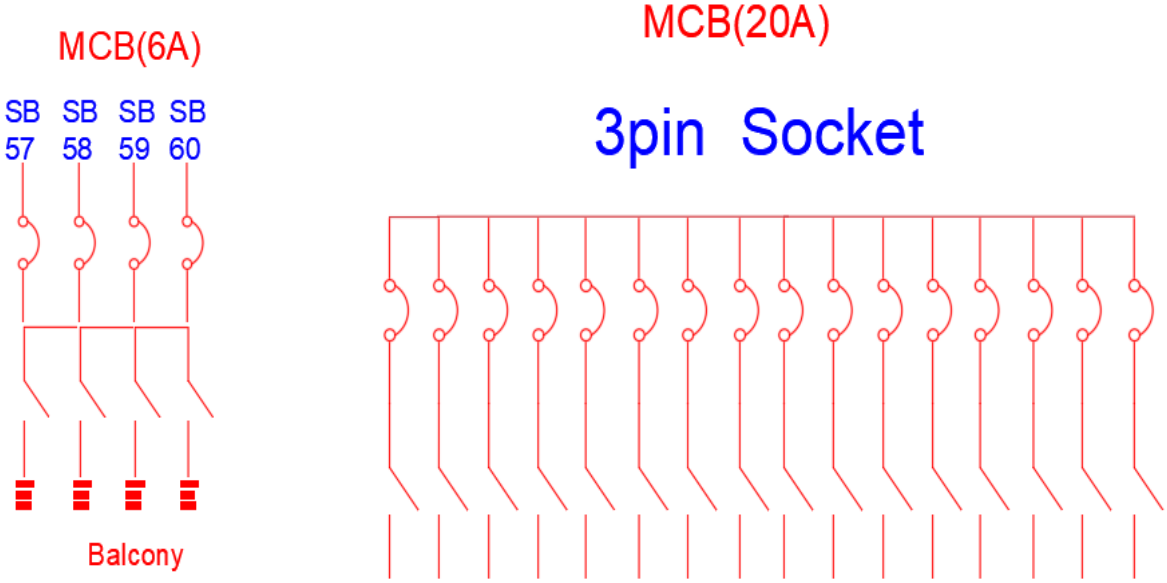
3pin Socket








SB52
MCB(6A)
SB53
MCB(6A)
SB54
MCB(6A)
SB55
MCB(6A)
SB56
MCB(6A)


MCB(6A)
SB 57
SB 58
SB 59
SB 60
Balcony
MCB(20A)
3pin Socket

Conduit Length Calculation

Specification as follows:

- **The Height of the Ceiling Fan from the Floor Level = 10.7 Feet.**
- **The Height of the LED tube light from the floor Level = 12 Feet.**
- **The Height of the Switch from the Floor Level = 4 Feet.**
- **The height of the MDB and SDB from the Floor Level = 6 Feet**
- **The Height of the 3-pin Socket from the floor Level = 1.5 Feet.**
- **The Height of the LED light from the Floor Level = 8 Feet.**

Conduit Length from Switchboard to Appliances:

Area/Space	Switchboard	Particular	Conduit Length	Total Length
ROOM-01	SB01	SB-01 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB02	SB-02 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-02	SB03	SB-03 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB04	SB-04 TO FAN	14’	

		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM03	SB05	SB-05 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB06	SB-06 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM04	SB07	SB-07 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB08	SB-08 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-05	SB09	SB-09 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	

	SB10	SB-10 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-06	SB11	SB-11 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB12	SB-12 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-07	SB13	SB-13 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB14	SB-14 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-08	SB15	SB-15 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	

		FAN TO LIGHT	8.5’	
	SB16	SB-16 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-09	SB17	SB-17 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB-18	SB-18 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-10	SB19	SB-19 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB20	SB-20 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-11	SB21	SB-21 TO FAN	15’	73’

		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB22	SB-22 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-12	SB23	SB-23 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB-24	SB-24 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
ROOM-13	SB25	SB-25 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB-26	SB-26 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	

ROOM-14	SB27	SB-27 TO FAN	15’	73’
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
	SB28	SB-28 TO FAN	14’	
		FAN TO LIGHT	8.5’	
		FAN TO LIGHT	8.5’	
		FAN TO FAN	10’	
BALCONY01	SB29	SB 29 TO BULB	5.5’	73’
BALCONY02	SB30	SB 30 TO BULB	5.5’	
BALCONY03	SB31	SB 31 TO BULB	5.5’	
BALCONY04	SB32	SB 32 TO BULB	5.5’	
BALCONY05	SB33	SB 33 TO BULB	5.5’	
BALCONY06	SB34	SB 34 TO BULB	5.5’	
BALCONY07	SB35	SB 35 TO BULB	5’	
BALCONY08	SB36	SB 36 TO BULB	5’	
BALCONY09	SB37	SB 37 TO BULB	5’	
BALCONY10	SB38	SB 38 TO BULB	5’	
BALCONY11	SB39	SB 39 TO BULB	5’	
BALCONY12	SB40	SB 40 TO BULB	5’	

BALCONY13	SB41	SB 41 TO BULB	5’	
BALCONY14	SB42	SB 42 TO BULB	5’	
CORRIDOR	SB43	SCL-03 TO SCL-04	28’	256’
		SCL-02 TO SCL-03	28’	
		SCL-01 TO SCL-02	28’	
		SB43 TO SCL-01	14’	
		SB43 TO STAIR BULB	30’	
	SB44	SCL-05 TO SCL-06	28’	
		SCL-06 TO SCL-07	28’	
		SCL-07 TO SCL-08	28’	
		SCL08 TO SB44	14’	
		SB44 TO STAIR BULB	30’	
BASIN AREA	SB45	SB45 TO BASIN AREA	15’	24.5’
		BASIN AREA TO SCL09	9.5’	
SHOWER	SB46	SCL09 TO SB46	15.5	20’
		SCL09 TO SCL10	4.5’	
WASHROOM	SB47	SB46 TO SB 47	1.75’	5.25’
	SB48	SB46 TO SB48	1.75’	
	SB49	SB48 TO	1.75’	

		SB49		
SHOWER	SB50	SB45 TO SB50	1.75’	3.50’
	SB51	SB50 TO SB51	1.75’	
BASIN AREA	SB52	SB52 TO BASIN AREA	15’	24.5’
		BASIN AREA TO SCL11	9.5’	
SHOWER	SB53	SCL11 TO SB53	15.5’	20’
		SCL11TO SCL12	4.5’	
WASHROOM	SB55	SB54 TO SB 55	1.75’	5.25’
	SB56	SB54TO SB56	1.75’	
	SB57	SB56 TO SB57	1.75’	
SHOWER	SB58	SB43 TO SB58	1.75’	3.5’
	SB59	SB58 TO SB59	1.75’	

Total Conduit Length = 1457.5’

Measurement of Conduit for Power Circuit & Switchboard:

Conduit calculation for Sub Distribution Board-01:

From	To	Area/Place	Conduit Length	Total Length
SDB01	SB01	ROOM-01	20.5’	98’
SB01	3 PIN SOCKET-1		37’	
3 PIN SOCKET-1	3 PIN SOCKET-2		11’	
SB01	3 PIN SOCKET-3		11.5’	
3 PIN SOCKET-3	3 PIN SOCKET-4		11’	
3 PIN SOCKET-4	SB02		7’	
SB02	SB03	BALCONY-01	5.5’	5.5’
SB01	SB04	ROOM02	19’	96.5’
SB04	3 PIN SOCKET-5		37’	
3 PIN SOCKET-5	3 PIN SOCKET-6		11’	
SB04	3 PIN SOCKET-7		11.5’	
3 PIN SOCKET-7	3 PIN SOCKET-8		11’	
3 PIN SOCKET-8	SB05		7’	
SB05	SB06	BALCONY-02	5.5’	5.5’
SB04	SB07	ROOM03	18’	95.5’
SB07	3 PIN SOCKET-9		37’	

3 PIN SOCKET-9	3 PIN SOCKET-10		11’	
SB07	3 PIN SOCKET-11		11.5’	
3 PIN SOCKET-11	3 PIN SOCKET-12		11’	
3 PIN SOCKET-12	SB08		7’	
SB08	SB09	BALCONY-03	5.5’	5.5’
SDB01	SB10	ROOM04	24’	100’
SB10	3 PIN SOCKET-13		21.5’	
SB10	3 PIN SOCKET-14		19.5’	
3 PIN SOCKET-14	3 PIN SOCKET-15		10.5’	
3 PIN SOCKET-13	3 PIN SOCKET-16		11’	
3 PIN SOCKET-16	SB11		13.5’	
SB11	SB12	BALCONY04	8.5’	8.5’
SB10	SB13	ROOM05	16.5’	92.5’
SB13	3 PIN SOCKET-17		21.5’	
SB13	3 PIN SOCKET-18		19.5’	
3 PIN SOCKET-18	3 PIN SOCKET-19		10.5’	
3 PIN SOCKET-17	3 PIN SOCKET-20		11’	
3 PIN SOCKET-20	SB14		13.5’	
SB14	SB15	BALCONY05	8.5’	8.5’
SDB01	SB16	STAIRCASE	36.5’	49.5’
SB16	SB17		13’	
SB17	SB18	WASHROOM	25’	85’
SB18	SB19		20’	

SB19	SB20		20’	
SB20	SB21		20’	
SB17	SB22	SHOWER	27.5’	67.5’
SB22	SB23		20’	
SB23	SB 24		20’	

Total conduit = 718’

Conduit calculation for Sub Distribution Board-02:

From	To	Area/Place	Conduit Length	Total Length
SDB01	SB01	ROOM-01	20.5’	98’
SB01	3 PIN SOCKET-1		37’	
3 PIN SOCKET-1	3 PIN SOCKET-2		11’	
SB01	3 PIN SOCKET-3		11.5’	
3 PIN SOCKET-3	3 PIN SOCKET-4		11’	
3 PIN SOCKET-4	SB02		7’	
SB02	SB03	BALCONY-01	5.5’	5.5’
SB01	SB04	ROOM02	19’	96.5’
SB04	3 PIN SOCKET-5		37’	
3 PIN SOCKET-5	3 PIN SOCKET-6		11’	
SB04	3 PIN SOCKET-7		11.5’	
3 PIN SOCKET-7	3 PIN SOCKET-8		11’	
3 PIN	SB05		7’	

SOCKET-8				
SB05	SB06	BALCONY-02	5.5’	5.5’
SB04	SB07	ROOM03	18’	95.5’
SB07	3 PIN SOCKET-9		37’	
3 PIN SOCKET-9	3 PIN SOCKET-10		11’	
SB07	3 PIN SOCKET-11		11.5’	
3 PIN SOCKET-11	3 PIN SOCKET-12		11’	
3 PIN SOCKET-12	SB08		7’	
SB08	SB09	BALCONY-03	5.5’	5.5’
SDB01	SB10	ROOM04	24’	100’
SB10	3 PIN SOCKET-13		21.5’	
SB10	3 PIN SOCKET-14		19.5’	
3 PIN SOCKET-14	3 PIN SOCKET-15		10.5’	
3 PIN SOCKET-13	3 PIN SOCKET-16		11’	
3 PIN SOCKET-16	SB11		13.5’	
SB11	SB12	BALCONY04	8.5’	8.5’
SB10	SB13	ROOM05	16.5’	92.5’
SB13	3 PIN SOCKET-17		21.5’	
SB13	3 PIN SOCKET-18		19.5’	
3 PIN SOCKET-18	3 PIN SOCKET-19		10.5’	
3 PIN SOCKET-17	3 PIN SOCKET-20		11’	
3 PIN SOCKET-20	SB14		13.5’	

SB14	SB15	BALCONY05	8.5’	8.5’
SDB01	SB16	STAIRCASE	36.5’	49.5’
SB16	SB17		13’	
SB17	SB18	WASHROOM	25’	85’
SB18	SB19		20’	
SB19	SB20		20’	
SB20	SB21		20’	
SB17	SB22	SHOWER	27.5’	67.5’
SB22	SB23		20’	
SB23	SB 24		20’	

Total conduit = 718’

Conduit calculation for Sub Distribution Board-03:

From	To	Area/Place	Conduit Length	Total Length
SDB03	SB01	ROOM11	17’	116.3’
SB01	3 PIN SOCKET-1		8.5’	
SB01	3 PIN SOCKET-2		33.75’	
3 PIN SOCKET-3	SB02		12.5’	
3 PIN SOCKET-1	3 PIN SOCKET-3		10.8’	
3 PIN SOCKET-2	3 PIN SOCKET-4		33.75’	
SB02	SB03	BALCONY 11	9’	9’
SB01	SB04		16’	115.3’

SB04	3 PIN SOCKET-5	ROOM12	8.5’	
SB04	3 PIN SOCKET-6		33.75’	
3 PIN SOCKET-8	SB05		12.5’	
3 PIN SOCKET-5	3 PIN SOCKET-7		10.8’	
3 PIN SOCKET-6	3 PIN SOCKET-8		33.75’	
SB05	SB06	BALCONY 12	9’	9’
SB04	SB07		16’	115.3’
SB07	3 PIN SOCKET-9	ROOM13	8.5’	
SB07	3 PIN SOCKET-10		33.75’	
3 PIN SOCKET-11	SB08		12.5’	
3 PIN SOCKET-9	3 PIN SOCKET-11		10.8’	
3 PIN SOCKET-10	3 PIN SOCKET-12		33.75’	
SB09	SB09	BALCONY 13	9’	9’
SB07	SB10		16’	115.3’
SB10	3 PIN SOCKET-13	ROOM 14	8.5’	
3 PIN SOCKET-13	3 PIN SOCKET-14		33.75’	
SB10	3 PIN SOCKET-15		12.5’	
3 PIN SOCKET-15	3 PIN SOCKET-16		10.8’	
3 PIN SOCKET-14	SB11		33.75’	
SB11	SB12	BALCONY14	9’	9’

Total conduit = 498.2’

Size Specification:

From switch board to light loads = 1.5 mm^2 (1/1.40mm),10A

From switch board to heavy loads (PO) = 2.5 mm^2 (1/1.78mm), 20A

From Sub-Distribution-Board to switch board = 2.5 mm^2 (1/1.78mm), 20A

Earthing wire size = 1.5 mm^2 (1/1.40mm), 10A (Single core)

Live Wire Calculation

Estimation of Wires for Lighting Circuit

Area/Space	Switchboard	Particular	Name of the Conductor	Wire Length	Total wire Length
ROOM-01	SB01	SB-01 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB01 TO LIGHT		23.5’	
		SB01 TO LIGHT		23.5’	
	SB02	SB-02 TO FAN		14’	
		SB-02 TO LIGHT		22.5’	
		SB-02 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-02	SB03	SB-03 TO	1 ×1.5RM	15’	131’

		FAN	(3-W)		
		SB-03 TO LIGHT		23.5’	
		SB-03 TO LIGHT		23.5’	
	SB04	SB-04 TO FAN		14’	
		SB-04 TO LIGHT		22.5’	
		SB-04 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM03	SB05	SB-05 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB-05 TO LIGHT		23.5’	
		SB-05 TO LIGHT		23.5’	
	SB06	SB-06 TO FAN		14’	
		SB-06 TO LIGHT		22.5’	
		SB-06 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM04	SB07	SB-07 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB-07 TO LIGHT		23.5’	
		SB-07 TO LIGHT		23.5’	
	SB08	SB-08 TO FAN		14’	
		SB08 TO LIGHT		22.5’	

		SB08 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-05	SB09	SB09 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB09 TO LIGHT		23.5’	
		SB09 TO LIGHT		23.5’	
	SB10	SB-10 TO FAN		14’	
		SB10 TO LIGHT		22.5’	
		SB10 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-06	SB11	SB-11 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB11 TO LIGHT		23.5’	
		SB11 TO LIGHT		23.5’	
	SB12	SB12 TO FAN		14’	
		SB12 TO LIGHT		22.5’	
		SB12 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-07	SB13	SB-13 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB13 TO LIGHT		23.5’	
		SB13 TO		23.5’	

		LIGHT			
	SB14	SB-14 TO FAN		14’	
		SB14 TO LIGHT		22.5’	
		SB14 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-08	SB15	SB-15 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB15 TO LIGHT		23.5’	
		SB15 TO LIGHT		23.5’	
	SB16	SB-16 TO FAN		14’	
		SB16 TO LIGHT		22.5’	
		SB16 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-09	SB17	SB-17 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB17 TO LIGHT		23.5’	
		SB17 TO LIGHT		23.5’	
	SB-18	SB-18 TO FAN		14’	
		SB-18 TO LIGHT		22.5’	
		SB-18 TO LIGHT		22.5’	
		FAN TO FAN		10’	

ROOM-10	SB19	SB-19 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB19 TO LIGHT		23.5’	
		SB19 TO LIGHT		23.5’	
	SB20	SB-20 TO FAN		14’	
		SB20 TO LIGHT		22.5’	
		SB20 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-11	SB21	SB21 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB21 TO LIGHT		23.5’	
		SB21 TO LIGHT		23.5’	
	SB22	SB-22 TO FAN		14’	
		SB22 TO LIGHT		22.5’	
		SB22 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-12	SB23	SB-23 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB23 TO LIGHT		23.5’	
		SB23 TO LIGHT		23.5’	
	SB-24	SB-24 TO FAN		14’	
		SB-24 TO		22.5’	

		LIGHT			
		SB-24 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-13	SB25	SB-25 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB25 TO LIGHT		23.5’	
		SB25 TO LIGHT		23.5’	
	SB-26	SB-26 TO FAN		14’	
		SB-26 TO LIGHT		22.5’	
		SB-26 TO LIGHT		22.5’	
		FAN TO FAN		10’	
ROOM-14	SB27	SB-27 TO FAN	1 ×1.5RM (3-W)	15’	131’
		SB27 TO LIGHT		23.5’	
		SB27 TO LIGHT		23.5’	
	SB28	SB-28 TO FAN		14’	
		SB28 TO LIGHT		22.5’	
		SB28 TO LIGHT		22.5’	
		FAN TO FAN		10’	
BALCONY01	SB29	SB 29 TO BULB	1 ×1.5RM (3-W)	5.5’	73’
BALCONY02	SB30	SB 29 TO BULB		5.5’	

BALCONY03	SB31	SB 30 TO BULB		5.5’	
BALCONY04	SB32	SB 31 TO BULB		5.5’	
BALCONY05	SB33	SB 32 TO BULB		5.5’	
BALCONY06	SB34	SB 33 TO BULB		5.5’	
BALCONY07	SB35	SB 34 TO BULB		5’	
BALCONY08	SB36	SB 35 TO BULB		5’	
BALCONY09	SB37	SB 36 TO BULB		5’	
BALCONY10	SB38	SB 37 TO BULB		5’	
BALCONY11	SB39	SB 38 TO SB19		5’	
BALCONY12	SB40	SB 39 TO BULB		5’	
BALCONY13	SB41	SB 40 TO BULB		5’	
BALCONY14	SB42	SB 41 TO BULB		5’	
CORRIDOR	SB43	SB43 TO SCL-04	1 ×1.5RM (3-W)	98’	508’
		SB43 TO SCL-03		70’	
		SB43 TO SCL-02		42’	
		SB43 TO SCL-01		14’	
		SB43 TO STAIR BULB		30’	
	SB44	SCL-05 TO SB44		98’	

		SCL-06 TO SB44		70’	
		SCL-07 TO SB44		42’	
		SCL08 TO SB44		14’	
		SB44 TO STAIR BULB		30’	
BASIN AREA	SB45	SB45 TO BASIN AREA	1 ×1.5RM (3-W)	15’	39.5’
		SB45TO SCL09		24.5’	
SHOWER	SB46	SCL09 TO SB46		15.5	35.5’
		SB46 TO SCL10		20’	
WASHROOM	SB48	SB47 TO SB 48		1.75’	5.25’
	SB49	SB47 TO SB49		1.75’	
	SB50	SB49 TO SB50		1.75’	
SHOWER	SB51	SB46 TO SB51		1.75’	3.50’
	SB52	SB51 TO SB52		1.75’	
BASIN AREA	SB53	SB53 TO BASIN AREA		15’	39.5’
		SB53 TO SCL11		24.5’	
SHOWER	SB54	SCL11 TO SB54		15.5’	35.5’
		SB54TO SCL12		20’	
WASHROOM	SB56	SB55 TO		1.75’	5.25’

		SB 56			
	SB57	SB55TO SB57		1.75’	
	SB58	SB57 TO SB58		1.75’	
SHOWER	SB59	SB54 TO SB59		1.75’	3.50’
	SB60	SB59 TO SB60		1.75’	

Wire calculation for Sub Distribution Board-01:

From	To	Area/Place	Name of the Conductor	Wire Length	Total Length
SDB01	SB01	ROOM-01	1 ×1.5RM (3-W)	20.5’	327’
	3 PIN SOCKET-1		1 × 2.5RM (3-W)	57.5’	
	3 PIN SOCKET-2			68.5’	
	3 PIN SOCKET-3			32’	
	3 PIN SOCKET-4			43’	
	SB02		1 ×1.5RM (3-W)	50’	
	SB03	BALCONY-01	1 ×1.5RM (3-W)	55.5’	

SDB-01	SB04	ROOM02	1 ×1.5RM (3-W)	39.5’	460’
	3 PIN SOCKET-5		1 × 2.5RM (3-W)	76.5’	
	3 PIN SOCKET-6			87.5’	
	3 PIN SOCKET-7			51’	
	3 PIN SOCKET-8			62’	
	SB05		1 ×1.5RM (3-W)	69’	
	SB06	BALCONY-02	1 ×1.5RM (3-W)	74.5’	
SDB01	SB07	ROOM03	1 ×1.5RM (3-W)	57.5’	514’
	3 PIN SOCKET-9		1 × 2.5RM (3-W)	94.5’	
	3 PIN SOCKET-10			105.5’	
	3 PIN SOCKET-11			69’	
	3 PIN SOCKET-12			80’	
	SB08		1 ×1.5RM (3-W)	87’	

	SB09	BALCONY-03	1 ×1.5RM (3-W)	92.5’	
SDB01	SB10	ROOM04	1 ×1.5RM (3-W)	24’	372’
	3 PIN SOCKET-13		1 × 2.5RM (3-W)	45.5’	
	3 PIN SOCKET-14			43.5’	
	3 PIN SOCKET-15			54’	
	3 PIN SOCKET-16			56.5’	
	SB11		1 ×1.5RM (3-W)	70’	
	SB12	BALCONY04	1 ×1.5RM (3-W)	78.5’	
SDB01	SB13	ROOM05	1 ×1.5RM (3-W)	40.5’	487.5’
	3 PIN SOCKET-17		1 × 2.5RM (3-W)	62’	
	3 PIN SOCKET-18			60’	
	3 PIN SOCKET-19			70.5’	

	3 PIN SOCKET-20			73’	
	SB14		1 ×1.5RM (3-W)	86.5’	
	SB15	BALCONY05	1 ×1.5RM (3-W)	95’	
SDB01	SB16	STAIRCASE	1 ×1.5RM (3-W)	36.5’	86’
	SB17			49.5’	
SDB01	SB18	WASHROOM		74.5’	418’
	SB19			94.5’	
	SB20			114.5’	
	SB21			134.5’	
SDB01	SB22	SHOWER		77’	291’
	SB23			97’	
	SB 24			117’	

Wire calculation for Sub Distribution Board-02:

From	To	Area/Place	Name of the Conductor	Wire Length	Total Length
SDB01	SB01	ROOM-01	1 ×1.5RM (3-W)	20.5’	327’
	3 PIN SOCKET-1		1 × 2.5RM (3-W)	57.5’	
	3 PIN SOCKET-2			68.5’	

	3 PIN SOCKET-3			32’	
	3 PIN SOCKET-4			43’	
	SB02		1 ×1.5RM (3-W)	50’	
	SB03	BALCONY-01	1 ×1.5RM (3-W)	55.5’	
SDB-01	SB04	ROOM02	1 ×1.5RM (3-W)	39.5’	460’
	3 PIN SOCKET-5		1 × 2.5RM (3-W)	76.5’	
	3 PIN SOCKET-6			87.5’	
	3 PIN SOCKET-7			51’	
	3 PIN SOCKET-8			62’	
	SB05		1 ×1.5RM (3-W)	69’	
	SB06	BALCONY-02	1 ×1.5RM (3-W)	74.5’	
SDB01	SB07	ROOM03	1 ×1.5RM (3-W)	57.5’	514’
	3 PIN SOCKET-		1 × 2.5RM (3-	94.5’	

	9		W)		
	3 PIN SOCKET-10			105.5’	
	3 PIN SOCKET-11			69’	
	3 PIN SOCKET-12			80’	
	SB08		1 ×1.5RM (3-W)	87’	
	SB09	BALCONY-03	1 ×1.5RM (3-W)	92.5’	
SDB01	SB10	ROOM04	1 ×1.5RM (3-W)	24’	372’
	3 PIN SOCKET-13		1 × 2.5RM (3-W)	45.5’	
	3 PIN SOCKET-14			43.5’	
	3 PIN SOCKET-15			54’	
	3 PIN SOCKET-16			56.5’	
	SB11		1 ×1.5RM (3-W)	70’	
	SB12	BALCONY04	1 ×1.5RM (3-W)	78.5’	
SDB01	SB13	ROOM05	1 ×1.5RM (3-	40.5’	487.5’

			W)		
	3 PIN SOCKET-17		1 × 2.5RM (3-W)	62’	
	3 PIN SOCKET-18			60’	
	3 PIN SOCKET-19			70.5’	
	3 PIN SOCKET-20			73’	
	SB14		1 ×1.5RM (3-W)	86.5’	
	SB15	BALCONY05	1 ×1.5RM (3-W)	95’	
SDB01	SB16	STAIRCASE	1 ×1.5RM (3-W)	36.5’	86’
	SB17			49.5’	
SDB01	SB18	WASHROOM		74.5’	418’
	SB19			94.5’	
	SB20			114.5’	
	SB21			134.5’	
SDB01	SB22	SHOWER		77’	291’
	SB23			97’	
	SB 24			117’	

Wire calculation for Sub Distribution Board-03:

From	To	Area/Place	Name of The Conductor	Wire Length	Total Length
SDB03	SB01	ROOM11	1 ×1.5RM (3-W)	17’	320.65’
	3 PIN SOCKET-1		1 × 2.5RM (3-W)	25.5’	
	3 PIN SOCKET-2			50.75’	
	SB02			48.8’	
	3 PIN SOCKET-3			36.3’	
	3 PIN SOCKET-4			84.5’	
	SB03	BALCONY 11	1 ×1.5RM (3-W)	57.8’	
SDB03	SB04	ROOM12	1 ×1.5RM (3-W)	33’	432.65’
	3 PIN SOCKET-5		1 × 2.5RM (3-W)	41.5’	
	3 PIN SOCKET-6			66.75’	
	SB05			64.8’	
	3 PIN SOCKET-7			52.3’	
	3 PIN SOCKET-8			100.5’	
	SB06	BALCONY 12	1 ×1.5RM (3-W)	73.8’	
SDB03	SB07	ROOM13	1 ×1.5RM	49’	544.65’

			(3-W)		
	3 PIN SOCKET-9		1 × 2.5RM (3-W)	57.5’	
	3 PIN SOCKET-10			82.75’	
	SB08			80.8’	
	3 PIN SOCKET-11			68.3’	
	3 PIN SOCKET-12			116.5’	
	SB09	BALCONY 13	1 ×1.5RM (3-W)	89.8’	
SDB03	SB10	ROOM 14	1 ×1.5RM (3-W)	65’	702.55’
	3 PIN SOCKET-13		1 × 2.5RM (3-W)	73.5’	
	3 PIN SOCKET-14			107.25’	
	3 PIN SOCKET-15			77.5’	
	3 PIN SOCKET-16			88.3’	
	SB11		1 ×1.5RM (3-W)	141’	
	SB12	BALCONY14	1 ×1.5RM (3-W)	150’	

Neutral Wire Calculation

Measurement of Neutral Wire for Power Circuit & Switchboard:

Neutral Wire for Sub Distribution Board-01:

From	To	Area/Place	Name Of the conductor	Wire Length	Total Length
SDB01	SB01	ROOM-01	1 ×1.5RM (3-W)	20.5’	98’
SB01	3 PIN SOCKET-1		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-1	3 PIN SOCKET-2			11’	
SB01	3 PIN SOCKET-3			11.5’	
3 PIN SOCKET-3	3 PIN SOCKET-4			11’	
3 PIN SOCKET-4	SB02		1 ×1.5RM (3-W)	7’	
SB02	SB03	BALCONY-01		5.5’	5.5’
SB01	SB04	ROOM02		19’	96.5’
SB04	3 PIN SOCKET-5		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-5	3 PIN SOCKET-6			11’	
SB04	3 PIN SOCKET-7			11.5’	
3 PIN SOCKET-7	3 PIN SOCKET-8			11’	
3 PIN SOCKET-8	SB05		1 ×1.5RM (3-W)	7’	
SB05	SB06	BALCONY-02		5.5’	5.5’
SB04	SB07			18’	

SB07	3 PIN SOCKET-9	ROOM03	1 × 2.5RM (3-W)	37’	95.5’
3 PIN SOCKET-9	3 PIN SOCKET-10			11’	
SB07	3 PIN SOCKET-11			11.5’	
3 PIN SOCKET-11	3 PIN SOCKET-12			11’	
3 PIN SOCKET-12	SB08		1 ×1.5RM (3-W)	7’	
SB08	SB09	BALCONY-03		5.5’	5.5’
SDB01	SB10	ROOM04		24’	100’
SB10	3 PIN SOCKET-13		1 × 2.5RM (3-W)	21.5’	
SB10	3 PIN SOCKET-14			19.5’	
3 PIN SOCKET-14	3 PIN SOCKET-15			10.5’	
3 PIN SOCKET-13	3 PIN SOCKET-16			11’	
3 PIN SOCKET-16	SB11		1 ×1.5RM (3-W)	13.5’	
SB11	SB12	BALCONY04		8.5’	8.5’
SB10	SB13	ROOM05		16.5’	92.5’
SB13	3 PIN SOCKET-17		1 × 2.5RM (3-W)	21.5’	
SB13	3 PIN SOCKET-18			19.5’	
3 PIN	3 PIN			10.5’	

SOCKET-18	SOCKET-19				
3 PIN SOCKET-17	3 PIN SOCKET-20			11’	
3 PIN SOCKET-20	SB14		1 ×1.5RM (3-W)	13.5’	
SB14	SB15	BALCONY05		8.5’	8.5’
SDB01	SB16	STAIRCASE		36.5’	49.5’
SB16	SB17			13’	
SB17	SB18	WASHROOM		25’	85’
SB18	SB19			20’	
SB19	SB20			20’	
SB20	SB21			20’	
SB17	SB22	SHOWER		27.5’	67.5’
SB22	SB23			20’	
SB23	SB 24			20’	

Neutral Wire for Sub Distribution Board-02:

From	To	Area/Place	Name Of the conductor	Wire Length	Total Length
SDB02	SB01	ROOM-01	1 ×1.5RM (3-W)	20.5’	98’
SB01	3 PIN SOCKET-1		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-1	3 PIN SOCKET-2			11’	
SB01	3 PIN SOCKET-3			11.5’	
3 PIN SOCKET-3	3 PIN SOCKET-4			11’	
3 PIN	SB02			7’	

SOCKET-4			1 ×1.5RM (3-W)		
SB02	SB03	BALCONY-01		5.5’	5.5’
SB01	SB04	ROOM02		19’	96.5’
SB04	3 PIN SOCKET-5		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-5	3 PIN SOCKET-6			11’	
SB04	3 PIN SOCKET-7			11.5’	
3 PIN SOCKET-7	3 PIN SOCKET-8			11’	
3 PIN SOCKET-8	SB05		1 ×1.5RM (3-W)	7’	
SB05	SB06	BALCONY-02		5.5’	5.5’
SB04	SB07	ROOM03		18’	95.5’
SB07	3 PIN SOCKET-9		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-9	3 PIN SOCKET-10			11’	
SB07	3 PIN SOCKET-11			11.5’	
3 PIN SOCKET-11	3 PIN SOCKET-12			11’	
3 PIN SOCKET-12	SB08		1 ×1.5RM (3-W)	7’	
SB08	SB09	BALCONY-03		5.5’	5.5’
SDB02	SB10	ROOM04		24’	100’
SB10	3 PIN SOCKET-13		1 × 2.5RM (3-W)	21.5’	
SB10	3 PIN SOCKET-14			19.5’	
3 PIN	3 PIN			10.5’	

SOCKET-14	SOCKET-15				
3 PIN SOCKET-13	3 PIN SOCKET-16			11’	
3 PIN SOCKET-16	SB11		1 ×1.5RM (3-W)	13.5’	
SB11	SB12	BALCONY04		8.5’	8.5’
SB10	SB13	ROOM05		16.5’	92.5’
SB13	3 PIN SOCKET-17		1 × 2.5RM (3-W)	21.5’	
SB13	3 PIN SOCKET-18			19.5’	
3 PIN SOCKET-18	3 PIN SOCKET-19			10.5’	
3 PIN SOCKET-17	3 PIN SOCKET-20			11’	
3 PIN SOCKET-20	SB14		1 ×1.5RM (3-W)	13.5’	
SB14	SB15	BALCONY05		8.5’	8.5’
SDB02	SB16	STAIRCASE		36.5’	49.5’
SB16	SB17			13’	
SB17	SB18	WASHROOM		25’	85’
SB18	SB19			20’	
SB19	SB20			20’	
SB20	SB21			20’	
SB17	SB22	SHOWER		27.5’	67.5’
SB22	SB23			20’	
SB23	SB 24			20’	

Neutral Wire calculation for Sub Distribution Board-03:

From	To	Area/Place	Name Of the conductor	Wire Length	Total Length
SDB03	SB01	ROOM11	1 ×1.5RM (3-W)	17’	116.3’
SB01	3 PIN SOCKET-1		1 × 2.5RM (3-W)	8.5’	
SB01	3 PIN SOCKET-2			33.75’	
3 PIN SOCKET-3	SB02		1 ×1.5RM (3-W)	12.5’	
3 PIN SOCKET-1	3 PIN SOCKET-3		1 × 2.5RM (3-W)	10.8’	
3 PIN SOCKET-2	3 PIN SOCKET-4			33.75’	
SB02	SB03	BALCONY 11	1 ×1.5RM (3-W)	9’	9’
SB01	SB04			16’	115.3’
SB04	3 PIN SOCKET-5	ROOM12	1 × 2.5RM (3-W)	8.5’	
SB04	3 PIN SOCKET-6			33.75’	
3 PIN SOCKET-8	SB05		1 ×1.5RM (3-W)	12.5’	
3 PIN SOCKET-5	3 PIN SOCKET-7		1 × 2.5RM (3-W)	10.8’	
3 PIN SOCKET-6	3 PIN SOCKET-8			33.75’	
SB05	SB06	BALCONY 12	1 ×1.5RM (3-W)	9’	9’
SB04	SB07			16’	115.3’

SB07	3 PIN SOCKET-9	ROOM13	1 × 2.5RM (3-W)	8.5’	
SB07	3 PIN SOCKET-10			33.75’	
3 PIN SOCKET-11	SB08		1 ×1.5RM (3-W)	12.5’	
3 PIN SOCKET-9	3 PIN SOCKET-11		1 × 2.5RM (3-W)	10.8’	
3 PIN SOCKET-10	3 PIN SOCKET-12			33.75’	
SB09	SB09	BALCONY 13	1 ×1.5RM (3-W)	9’	9’
SB07	SB10			16’	115.3’
SB10	3 PIN SOCKET-13	ROOM 14	1 × 2.5RM (3-W)	8.5’	
3 PIN SOCKET-13	3 PIN SOCKET-14			33.75’	
SB10	3 PIN SOCKET-15			12.5’	
3 PIN SOCKET-15	3 PIN SOCKET-16			10.8’	
3 PIN SOCKET-14	SB11		1 ×1.5RM (3-W)	33.75’	
SB11	SB12	BALCONY14		9’	9’

Neutral Wire Length from Switchboard to Appliances:

Area/Space	Switchboard	Particular	Name of the Conductor	Neutral Wire Length	Total Length
ROOM-01	SB01	SB-01 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB02	SB-02 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-02	SB03	SB-03 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB04	SB-04 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM03	SB05	SB-05 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	

		FAN TO LIGHT		8.5’	
	SB06	SB-06 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM04	SB07	SB-07 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB08	SB-08 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-05	SB09	SB-09 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB10	SB-10 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO		10’	

		FAN			
ROOM-06	SB11	SB-11 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB12	SB-12 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-07	SB13	SB-13 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB14	SB-14 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-08	SB15	SB-15 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB16	SB-16 TO FAN		14’	

		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-09	SB17	SB-17 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB-18	SB-18 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-10	SB19	SB-19 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB20	SB-20 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-11	SB21	SB-21 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO		8.5’	

		LIGHT			
		FAN TO LIGHT		8.5’	
	SB22	SB-22 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-12	SB23	SB-23 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB-24	SB-24 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
ROOM-13	SB25	SB-25 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB-26	SB-26 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	

		FAN TO FAN		10’	
ROOM-14	SB27	SB-27 TO FAN	1 ×1.5RM (3-W)	15’	73’
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
	SB28	SB-28 TO FAN		14’	
		FAN TO LIGHT		8.5’	
		FAN TO LIGHT		8.5’	
		FAN TO FAN		10’	
BALCONY01	SB29	SB 29 TO BULB	1 ×1.5RM (3-W)	5.5’	73’
BALCONY02	SB30	SB 30 TO BULB		5.5’	
BALCONY03	SB31	SB 31 TO BULB		5.5’	
BALCONY04	SB32	SB 32 TO BULB		5.5’	
BALCONY05	SB33	SB 33 TO BULB		5.5’	
BALCONY06	SB34	SB 34 TO BULB		5.5’	
BALCONY07	SB35	SB 35 TO BULB		5’	
BALCONY08	SB36	SB 36 TO BULB		5’	
BALCONY09	SB37	SB 37 TO BULB		5’	
BALCONY10	SB38	SB 38 TO BULB		5’	
BALCONY11	SB39	SB 39 TO		5’	

		BULB			
BALCONY12	SB40	SB 40 TO BULB		5’	
BALCONY13	SB41	SB 41 TO BULB		5’	
BALCONY14	SB42	SB 42 TO BULB		5’	
CORRIDOR	SB43	SCL-03 TO SCL-04	1 ×1.5RM (3-W)	28’	256’
		SCL-02 TO SCL-03		28’	
		SCL-01 TO SCL-02		28’	
		SB43 TO SCL-01		14’	
		SB43 TO STAIR BULB		30’	
	SB44	SCL-05 TO SCL-06		28’	
		SCL-06 TO SCL-07		28’	
		SCL-07 TO SCL-08		28’	
		SCL08 TO SB44		14’	
		SB44 TO STAIR BULB		30’	
BASIN AREA	SB45	SB45 TO BASIN AREA	1 ×1.5RM (3-W)	15’	24.5’
		BASIN AREA TO SCL09		9.5’	
SHOWER	SB46	SCL09 TO SB46	1 ×1.5RM (3-W)	15.5	20’
		SCL09 TO		4.5’	

		SCL10			
WASHROOM	SB47	SB46 TO SB 47		1.75’	5.25’
	SB48	SB46 TO SB48		1.75’	
	SB49	SB48 TO SB49		1.75’	
SHOWER	SB50	SB45 TO SB50		1.75’	3.50’
	SB51	SB50 TO SB51		1.75’	
BASIN AREA	SB52	SB52 TO BASIN AREA		15’	24.5’
		BASIN AREA TO SCL11		9.5’	
SHOWER	SB53	SCL11 TO SB53		15.5’	20’
		SCL11TO SCL12		4.5’	
WASHROOM	SB55	SB54 TO SB 55		1.75’	5.25’
	SB56	SB54TO SB56		1.75’	
	SB57	SB56 TO SB57		1.75’	
SHOWER	SB58	SB43 TO SB58		1.75’	3.5’
	SB59	SB58 TO SB59		1.75’	

ECC wire Calculation

Measurement of ECC for Power Circuit & Switchboard:

ECC for Sub Distribution Board-01:

From	To	Area/Place	Name Of the conductor	Wire Length	Total Length
SDB01	SB01	ROOM-01	1 ×1.5RM (3-W)	20.5’	98’
SB01	3 PIN SOCKET-1		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-1	3 PIN SOCKET-2			11’	
SB01	3 PIN SOCKET-3			11.5’	
3 PIN SOCKET-3	3 PIN SOCKET-4			11’	
3 PIN SOCKET-4	SB02		1 ×1.5RM (3-W)	7’	
SB02	SB03	BALCONY-01		5.5’	5.5’
SB01	SB04	ROOM02		19’	96.5’
SB04	3 PIN SOCKET-5		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-5	3 PIN SOCKET-6			11’	
SB04	3 PIN SOCKET-7			11.5’	
3 PIN SOCKET-7	3 PIN SOCKET-8			11’	
3 PIN SOCKET-8	SB05		1 ×1.5RM (3-W)	7’	
SB05	SB06	BALCONY-02		5.5’	5.5’

SB04	SB07	ROOM03		18’	95.5’
SB07	3 PIN SOCKET-9		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-9	3 PIN SOCKET-10			11’	
SB07	3 PIN SOCKET-11			11.5’	
3 PIN SOCKET-11	3 PIN SOCKET-12			11’	
3 PIN SOCKET-12	SB08		1 ×1.5RM (3-W)	7’	
SB08	SB09	BALCONY-03		5.5’	5.5’
SDB01	SB10	ROOM04		24’	100’
SB10	3 PIN SOCKET-13		1 × 2.5RM (3-W)	21.5’	
SB10	3 PIN SOCKET-14			19.5’	
3 PIN SOCKET-14	3 PIN SOCKET-15			10.5’	
3 PIN SOCKET-13	3 PIN SOCKET-16			11’	
3 PIN SOCKET-16	SB11		1 ×1.5RM (3-W)	13.5’	
SB11	SB12	BALCONY04		8.5’	8.5’
SB10	SB13	ROOM05		16.5’	92.5’
SB13	3 PIN SOCKET-17		1 × 2.5RM (3-W)	21.5’	
SB13	3 PIN SOCKET-18			19.5’	

3 PIN SOCKET-18	3 PIN SOCKET-19			10.5’	
3 PIN SOCKET-17	3 PIN SOCKET-20			11’	
3 PIN SOCKET-20	SB14		1 ×1.5RM (3-W)	13.5’	
SB14	SB15	BALCONY05		8.5’	8.5’
SDB01	SB16	STAIRCASE		36.5’	49.5’
SB16	SB17			13’	
SB17	SB18	WASHROOM		25’	85’
SB18	SB19			20’	
SB19	SB20			20’	
SB20	SB21			20’	
SB17	SB22	SHOWER		27.5’	67.5’
SB22	SB23			20’	
SB23	SB 24			20’	

ECC for Sub Distribution Board-02:

From	To	Area/Place	Name Of the conductor	Wire Length	Total Length
SDB02	SB01	ROOM-01	1 ×1.5RM (3-W)	20.5’	**98**’
SB01	3 PIN SOCKET-1		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-1	3 PIN SOCKET-2			11’	
SB01	3 PIN SOCKET-3			11.5’	
3 PIN SOCKET-3	3 PIN SOCKET-4			11’	

3 PIN SOCKET-4	SB02		1 ×1.5RM (3-W)	7’	
SB02	SB03	BALCONY-01		5.5’	5.5’
SB01	SB04	ROOM02		19’	96.5’
SB04	3 PIN SOCKET-5		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-5	3 PIN SOCKET-6			11’	
SB04	3 PIN SOCKET-7			11.5’	
3 PIN SOCKET-7	3 PIN SOCKET-8			11’	
3 PIN SOCKET-8	SB05		1 ×1.5RM (3-W)	7’	
SB05	SB06	BALCONY-02		5.5’	5.5’
SB04	SB07	ROOM03		18’	95.5’
SB07	3 PIN SOCKET-9		1 × 2.5RM (3-W)	37’	
3 PIN SOCKET-9	3 PIN SOCKET-10			11’	
SB07	3 PIN SOCKET-11			11.5’	
3 PIN SOCKET-11	3 PIN SOCKET-12			11’	
3 PIN SOCKET-12	SB08		1 ×1.5RM (3-W)	7’	
SB08	SB09	BALCONY-03		5.5’	5.5’
SDB02	SB10	ROOM04		24’	100’
SB10	3 PIN SOCKET-13		1 × 2.5RM (3-W)	21.5’	
SB10	3 PIN SOCKET-14			19.5’	

3 PIN SOCKET-14	3 PIN SOCKET-15			10.5’	
3 PIN SOCKET-13	3 PIN SOCKET-16			11’	
3 PIN SOCKET-16	SB11		1 ×1.5RM (3-W)	13.5’	
SB11	SB12	BALCONY04		8.5’	8.5’
SB10	SB13	ROOM05		16.5’	92.5’
SB13	3 PIN SOCKET-17		1 × 2.5RM (3-W)	21.5’	
SB13	3 PIN SOCKET-18			19.5’	
3 PIN SOCKET-18	3 PIN SOCKET-19			10.5’	
3 PIN SOCKET-17	3 PIN SOCKET-20			11’	
3 PIN SOCKET-20	SB14		1 ×1.5RM (3-W)	13.5’	
SB14	SB15	BALCONY05		8.5’	8.5’
SDB02	SB16	STAIRCASE		36.5’	49.5’
SB16	SB17			13’	
SB17	SB18	WASHROOM		25’	85’
SB18	SB19			20’	
SB19	SB20			20’	
SB20	SB21			20’	
SB17	SB22	SHOWER		27.5’	67.5’
SB22	SB23			20’	
SB23	SB 24			20’	

ECC for Sub Distribution Board-03:

From	To	Area/Place	Name Of the conductor	Conduit Length	Total Length
SDB03	SB01	ROOM11	1 ×1.5RM (3-W)	17’	116.3’
SB01	3 PIN SOCKET-1		1 × 2.5RM (3-W)	8.5’	
SB01	3 PIN SOCKET-2			33.75’	
3 PIN SOCKET-3	SB02		1 ×1.5RM (3-W)	12.5’	
3 PIN SOCKET-1	3 PIN SOCKET-3		1 × 2.5RM (3-W)	10.8’	
3 PIN SOCKET-2	3 PIN SOCKET-4			33.75’	
SB02	SB03	BALCONY 11	1 ×1.5RM (3-W)	9’	9’
SB01	SB04			16’	115.3’
SB04	3 PIN SOCKET-5	ROOM12	1 × 2.5RM (3-W)	8.5’	
SB04	3 PIN SOCKET-6			33.75’	
3 PIN SOCKET-8	SB05		1 ×1.5RM (3-W)	12.5’	
3 PIN SOCKET-5	3 PIN SOCKET-7		1 × 2.5RM (3-W)	10.8’	
3 PIN SOCKET-6	3 PIN SOCKET-8			33.75’	
SB05	SB06	BALCONY 12	1 ×1.5RM (3-W)	9’	9’
SB04	SB07			16’	115.3’

SB07	3 PIN SOCKET-9	ROOM13	1 × 2.5RM (3-W)	8.5’	
SB07	3 PIN SOCKET-10			33.75’	
3 PIN SOCKET-11	SB08		1 ×1.5RM (3-W)	12.5’	
3 PIN SOCKET-9	3 PIN SOCKET-11		1 × 2.5RM (3-W)	10.8’	
3 PIN SOCKET-10	3 PIN SOCKET-12			33.75’	
SB09	SB09	BALCONY 13	1 ×1.5RM (3-W)	9’	9’
SB07	SB10			16’	115.3’
SB10	3 PIN SOCKET-13	ROOM 14	1 × 2.5RM (3-W)	8.5’	
3 PIN SOCKET-13	3 PIN SOCKET-14			33.75’	
SB10	3 PIN SOCKET-15			12.5’	
3 PIN SOCKET-15	3 PIN SOCKET-16			10.8’	
3 PIN SOCKET-14	SB11		1 ×1.5RM (3-W)	33.75’	
SB11	SB12	BALCONY14		9’	9’

For Main Distribution Board,

CONDUIT CALCULATION:

From	To	Conduit Length	Total Length
MDB	SDB01	86.5’	216.5’
	SDB02	80’	
	SDB03	50’	

Live Wire Calculation for MDB:

From	To	Name of The Conductor	Wire Length	Total Wire Length
MDB	SDB01	3 x 50 RM BYA-FR Copper Cable	86.5’	216.5’
	SDB02		80’	
	SDB03		50’	

Neutral Wire Calculation for MDB:

From	To	Name of The Conductor	Wire Length	Total Wire Length
MDB	SDB01	1 x 50 RM BYA-FR Copper Cable	86.5’	216.5’
	SDB02		80’	
	SDB03		50’	

ECC Wire Calculation for MDB:

From	To	Name of The Conductor	Wire Length	Total Wire Length
MDB	SDB01	1 x 25 RM BYA-FR Copper Cable	86.5’	216.5’
	SDB02		80’	
	SDB03		50’	

Estimation of Cost

Sl no.	Materials Used		Quantity	Rate (Tk)	Cost (Tk)
1	RFL uPVC Conduit		287 pcs (3/4”)	55/-	15785/-
			123 pcs (1”)	78/-	9594/-
			26 pcs (1.5”)	200/-	5200/-
2	Conduit fittings	Circular Box (2 way)	95	18/-	1710/-
3		Circular Box (3 way)	20	18/-	360/-
4		Circular Box (4 way)	30	22/-	660/-
5		Junction Box	45	45/-	2025/-
6		Outlet Box	56	30/-	1680/-
7		Rectangular Box	30	10/-	300/-
8		Inspection Box	30	40/-	1200/-
9		Elbow	115	15/-	1725/-
10		Coupler	200	12/-	2400/-
11		Tee	35	18/-	630/-
12	Concealed switch box		60	53/-	3180/-
13	G.I wire (pushing)		20Kg	150/-	3000/-
14	Insulation tape		100	25/-	2500/-
15	6A Single Pole MCB		60	216/-	12960/-
16	20A Single Pole MCB		56	216/-	12096/-

17	Main Distribution Board	200A 3-Phase MCCB	1		15500/-
		Panel Box, Copper Busbar, Accessories and labor cost	-		22000/-
18	Sub Distribution Board	63A 3-Phase MCB	2	7000/-	14000/-
		Panel Box, Copper Busbar, Accessories and labor cost	2	8500/-	17000/-
		50A 3-Phase MCB	1	2500	2500
		Panel Box, Copper Busbar, Accessories and labor cost		11500	11500
19	Live Wire (Red)	1 ×1.5RM (3-W)	6730’	14.63/- (per feet)	98462/-
		1 ×2.5RM (3-W)	3908.1’	18.12/ -(per feet)	70824/-
		3 x 50 RM (3-W) (BYA-FR) BRB Cables	650’	378/ -(per feet)	245700/-
20		1 ×1.5RM (3-W)	2350’	14.63/- (per feet)	34377/-

	Neutral wire (Black)	1 × 2.5 RM (7-W)	969’	18.12/- (per feet)	17558/-
		1 x 50 RM (3-W) (BYA-FR) BRB Cables	216.5’	378/- (per feet)	81837/-
21	Earthing Wire (Green)	1 ×1.5RM (3-W)	2350’	14.63/- (per feet)	34377/-
		1 × 2.5 RM (7-W)	969’	18.12/- (per feet)	17558/-
		1 x 25 RM (3-W) (BYA-FR) BRB Cables	216.5’	183/- (per feet)	39600/-
22	Surface Celling Light(50W)		8	1230/-	9840/-
23	Surface Celling Light(20W)		4	560/-	3440/-
24	Lube light (20W)		2	525/-	1050/-
25	LED bulb(12W)		56	330/-	18480/-
26	LED bulb(3W)		14	157/-	2198/-
27	Solar LED bulb(6W)		14	280/-	3920/-
28	LED bulb(5W)		2	248/-	496/-
29	LED bulb(50W)		2	1295/-	2590/-
30	Celling Fan		28	3390/-	94920/-
31	3 pin Socket (13A)		56	290/-	16240/-
32	5-gang switch		2	162/-	324/-
33	3-gang switch		30	139/-	4170/-
34	1-gang switch		28	110/-	3080/-
35	Screw (3/4”)		20 dozen	30/-	600/-
36	Screw (1”)		15 dozen	40/-	600/-
37	Royal plug		35 dozen	20/-	700/-
38	Labor Cost				75000/-

39	Miscellaneous			10000/-
Total				**10,47,446/-**

Discussion:

Upon finalizing all the design steps and performing the necessary technical calculations, the total optimum cost for the electrical wiring of the ground floor has been estimated at **10,47,446/-Tk**. This cost evaluation ensures an ideal balance between quality, safety, and efficiency. To maintain accuracy, all material prices were sourced from trusted and reputable suppliers.

All major electrical loads, switches, and protective devices were selected from Click Electronics Ltd., a reliable brand known for consistent performance and compliance with safety standards. The wiring cables—one of the most critical components of the entire system—were chosen from BRB Cables Ltd., EASTERN Cables Ltd., and WALTON, ensuring durability, proper insulation quality, and adherence to national electrical codes.

Ceiling fans were sourced from Walton Electronics, selected for their energy efficiency and long-term reliability. Additional necessary materials, such as conduit fittings, switches, sockets, junction boxes, wiring accessories, and labor charges, were calculated based on prevailing local market rates to ensure economic feasibility.

This comprehensive cost structure not only reflects material and installation expenses but also highlights a design approach focused on sustainability, safety, and enhanced user comfort. The combination of high-quality components and optimized layout ensures long-term reliability, reduced maintenance, and improved overall convenience. Ultimately, this investment contributes to creating a more comfortable, visually appealing, and functionally efficient environment for the occupants.

Conclusion:

This project has been undertaken as a comprehensive study of the electrical wiring system and its associated cost estimation for a building. Electrical wiring is a fundamental part of any construction, ensuring the safe and efficient distribution of electricity throughout the structure. A properly planned wiring system is essential for the smooth operation of electrical devices, as well as for the safety, functionality, and comfort of the building's occupants. Therefore, it is crucial for an Electrical Engineer to have a thorough understanding of wiring systems and their integration with architectural design.

In this project, the electrical wiring has been designed in alignment with the architectural plans to ensure practical and efficient installation. The work involves determining the placement of conduits, switches, sockets, lighting fixtures, and other electrical accessories, while calculating the required load, current, and distribution system for optimal performance. The project also emphasizes cost efficiency, aiming to provide maximum functionality and satisfaction at the minimum possible expense.

By carefully analyzing requirements, estimating material quantities, selecting quality components, and calculating costs, the project ensures that the final installation is safe, reliable**,** durable, and economically viable. This approach highlights the role of the Electrical Engineer in balancing technical expertise, resource management, and practical application to deliver a wiring system that is efficient, cost-effective, and suitable for modern building needs.

Reference:

[1] Bangladesh National Building Code (BNBC), *Electrical and Electronic Engineering Services for Buildings*, Housing and Building Research Institute (HBRI), Government of Bangladesh, 2020 Edition.

[2] S. L. Uppal, *Electrical Wiring, Estimating and Costing*, Khanna Publishers, New Delhi, India.

[3] BBS Cables Ltd., *Company Profile and Product Information*.
Available: Products Details -BBS Cable
[4] BRB Cables Industries Ltd., *Cable Price List*.
Available: Indoor / Domestic Cables – BRB Cable Industries Ltd

[5] Super Star Group, *LED Bracket Tube Light Product Information*.
Available: Super Star LED Bracket Tube Light

[6] BD Electrical Website, *Product Details*.
Available: Industrial Electrical Equipment for Sale | BD Electrical

[7] National Polymer (nationalpolymer.net)
Available: uPVC Pipe & Fittings | National Polymer Group | Top Pipe Company in Bangladesh

[8] Walton Hi-Tech Industries Ltd., *Switch and Socket Accessories*.
Available: Accessories

[9] Concealed Switch Box. Available: **https://ssgeshop.com/product/concealed-metal-switch-box-gangsage**

[10] Impress Electronics Ltd., *ABB MCCB Price List in Bangladesh*.
Available ABB Molded Case Circuit Breaker (MCCB) pricelist Bangladesh | Impress Corporation Limited
[11] Lighting BD Limited (Osram, Phillips). Details in: LED Bulb

[12] Click Electronics Ltd. Buy click led Online at Best Price in Bangladesh - Daraz.com.bd

[13] RFL Bangladesh. Details: Vigo Electric Conduit Pipe | RFL Group